

# INTERNATIONAL STANDARD

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**Cable networks for television signals, sound signals and interactive services –  
Part 13: Optical systems for broadcast signal transmissions**



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Part 13: Optical systems for broadcast signal transmissions**

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ELECTROTECHNICAL  
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**CABLE NETWORKS FOR TELEVISION SIGNALS,  
SOUND SIGNALS AND INTERACTIVE SERVICES –****Part 13: Optical systems for broadcast signal transmissions**

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International Standard IEC 60728-13 has been prepared by technical area 5: Cable networks for television signals, sound signals and interactive services, of IEC technical committee 100: Audio, video and multimedia systems and equipment.

The text of this standard is based on the following documents:

| FDIS          | Report on voting |
|---------------|------------------|
| 100/1623/FDIS | 100/1646/RVD     |

Full information on the voting for the approval of this standard can be found in the report on voting indicated in the above table.

This publication has been drafted in accordance with the ISO/IEC Directives, Part 2.

The list of all the parts of the IEC 60728 series, under the general title *Cable networks for television signals, sound signals and interactive services*, can be found on the IEC website.

The committee has decided that the contents of this publication will remain unchanged until the maintenance result date indicated on the IEC web site under "<http://webstore.iec.ch>" in the data related to the specific publication. At this date, the publication will be

- reconfirmed,
- withdrawn,
- replaced by a revised edition, or
- amended.

A bilingual version of this publication may be issued at a later date.

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## INTRODUCTION

Standards of the IEC 60728 series deal with cable networks including equipment and associated methods of measurement for headend reception, processing and distribution of television signals, sound signals and their associated data signals and for processing, interfacing and transmitting all kinds of signals for interactive services using all applicable transmission media.

This includes

- CATV<sup>1</sup>-networks;
- MATV-networks and SMATV-networks;
- individual receiving networks;

and all kinds of equipment, systems and installations installed in such networks.

The extent of this standardization work is from the antennas and/or special signal source inputs to the headend or other interface points to the network up to the terminal input.

The standardization of any user terminals (i.e., tuners, receivers, decoders, multimedia terminals, etc.) as well as of any coaxial, balanced and optical cables and accessories thereof is excluded.

---

<sup>1</sup> This word encompasses the HFC (Hybrid Fibre Cable) networks used nowadays to provide telecommunications services, voice, data, audio and video both broadcast and narrowcast.

# **CABLE NETWORKS FOR TELEVISION SIGNALS, SOUND SIGNALS AND INTERACTIVE SERVICES –**

## **Part 13: Optical systems for broadcast signal transmissions**

### **1 Scope**

This part of IEC 60728 is applicable to optical transmission system for broadcast signal transmission that consists of a head-end equipment, optical transmission lines, in-house wirings and a system outlet. The system is primarily intended for television and sound signals using analogue and/or digital transmission technology. This standard specifies the basic system parameters and methods of measurement for optical distribution system having a system outlet in order to assess the system performance and its performance limits.

The purpose of this part of IEC 60728 is to describe the system specification of FTTH (fibre to the home) network for broadcast signal transmission. This standard is also applicable to the broadcast signal transmission using telecommunication network if it satisfies the optical portion of this standard. This standard describes RF transmission for broadcast and narrowcast (limited area distribution of broadcast) signals over FTTH, and introduces xPON system as a physical layer media. The detailed description of physical layer is out of the scope of this standard. The scope is limited to RF signal transmission over FTTH, thus, it does not include IP transport technologies, such as IP Multicast and associate protocols. Some interference descriptions between telecommunication system and broadcast system addressed in Clause 7 and Annex D should be referred to for detailed explanations. Annex A describes actual service systems with design consideration based on this standard. Annex B gives an overview of the optical transmission systems applicable for broadcast signal transmission.

### **2 Normative references**

The following referenced documents are indispensable for the application of this document. For dated references, only the edition cited applies. For undated references, the latest edition of the referenced document (including any amendments) applies.

IEC 60068-1:1988, *Environmental testing – Part 1: General and guidance*

IEC 60728-1:2007, *Cable networks for television signals, sound signals and interactive services – Part 1: System performance of forward paths*

IEC 60728-6:2003, *Cable networks for television signals, sound signals and interactive services – Part 6: Optical equipment*

IEC/TR 60728-6-1:2006, *Cable networks for television signals, sound signals and interactive services – Part 6-1: System guidelines for analogue optical transmission systems*

IEC 60825-1, *Safety of laser products – Part 1: Equipment classification and requirements*

IEC 60825-2, *Safety of laser products – Part 2: Safety of optical fibre communication systems (OFCS)*

IEC 60825-12, *Safety of laser products – Part 12: Safety of free space optical communication systems used for transmission of information*

IEC 61291-1:2006, *Optical amplifiers – Part 1: Generic specification*

IEC 61755-1:2005, *Fibre optic connector optical interfaces – Part 1: Optical interfaces for single mode non-dispersion shifted fibres – General and guidance*

IEC 61930:1998, *Fibre optic graphical symbology*

IEC 61931:1998, *Fibre optic – Terminology*

ITU-T Recommendation G.692, *Optical interfaces for multichannel systems with optical amplifiers*

ITU-T Recommendation G.694.2, *Spectral grids for WDM applications: CWDM wavelength grid*

### **3 Terms, definitions, symbols and abbreviations**

#### **3.1 Terms and definitions**

For the purposes of this document, the following terms and definitions apply.

##### **3.1.1**

##### **optical transmitting unit**

##### **optical transmitter**

transmit fibre optic terminal device accepting at its input port an electrical signal and providing at its output port an optical carrier modulated by that input signal

[IEC 61931, definition 2.9.6]

NOTE 1 For the purposes of this document, optical transmitters may have more than one input port accepting electrical RF signals.

NOTE 2 This piece of equipment amplifies frequency multiplexed electrical signals and converts these electrical signals into optical signals. The optical wavelength is a 1 500 nm band ( $1\,550 \pm 10$  nm in 1 530 nm to 1 625 nm region).

##### **3.1.2**

##### **optical receiving unit**

##### **optical receiver**

receive fibre optic terminal device accepting at its input port a modulated optical carrier, and providing at its output port the corresponding demodulated electrical signal (with the associated clock, if digital)

[IEC 61931, definition 2.9.7]

NOTE For the purposes of this document, optical receivers may have more than one output port providing electrical RF signals.

##### **3.1.3**

##### **optical amplifier**

optical waveguide device containing a suitably pumped, active medium which is able to amplify an optical signal

[IEC 61931, definition 2.7.75]

NOTE 1 In this document, Erbium Doped Fibre Amplifier (EDFA) is used for amplification in the 1 550 nm band.

NOTE 2 There are several methods based on wavelength to be used for amplification. The term “Erbium Doped Fibre Amplifier (EDFA)” is the synonym of optical amplifier in this document.

### 3.1.4

#### **fibre optic branching device** **optical fibre coupler** **splitter**

optical fibre device, possessing three or more optical ports, which shares optical power among its ports in a predetermined fashion, at the same wavelength or wavelengths, without wavelength conversion

NOTE The ports may be connected to fibres, detectors, etc.

[IEC 61931, definition 2.6.21, modified]

### 3.1.5

#### **multiplexing device** **WDM device**

wavelength selective branching device (used in WDM transmission systems) in which optical signals can be transferred between two predetermined ports, depending on the wavelength of the signal

[IEC 61931, definition 2.6.51]

### 3.1.6

#### **optical modulation index**

optical modulation index of  $k^{\text{th}}$  RF carrier,  $m_k$  is defined as

$$m_k = \frac{\phi_h - \phi_l}{\phi_h + \phi_l}$$

total optical modulation index,  $M$  is defined as

$$M = \sqrt{\sum_{k=1}^K m_k^2}$$

where

$\phi_h$  is the highest and

$\phi_l$  is the lowest instantaneous optical power of the intensity modulated optical signal,

$K$  is the total number of RF carriers and

$M$  is the total optical modulation index.

NOTE This term is mainly used for analogue systems.

[IEC 60728-6, definition 3.1.10, modified]

### 3.1.7

#### **noise figure**

decrease of the signal-to-noise ratio (SNR), at the output of an optical detector with unitary quantum efficiency and zero excess noise, due to the propagation of a shot noise-limited signal through the optical fibre amplifier, expressed in dB

[IEC 61291-1, definition 3.2.38]

NOTE The noise figure of optical amplifiers depends on the optical input power and on the wavelength used.

**3.1.8****relative intensity noise****RIN**

ratio of the mean square of the intensity fluctuations in the optical power of a light source to the square of the mean of the optical output power

NOTE 1 The RIN is usually expressed in dB(Hz<sup>-1</sup>) resulting in negative values.

[IEC 60728-6, definition 3.1.12, modified]

NOTE 2 The value of RIN can also be calculated from the results of a carrier-to-noise measurement for the system.

**3.1.9****responsivity**

ratio of an optical detector's electrical output to its optical input at a given wavelength

NOTE 1 The responsivity is generally expressed in Ampere per Watt or Volt per Watt of incident radiant power.

NOTE 2 Sensitivity is sometimes used as an imprecise synonym for responsivity.

NOTE 3 The wavelength interval around the given wavelength may be specified.

[IEC 60728-6, definition 3.1.15]

**3.1.10****wavelength**

distance covered in a period by the wavefront of a harmonic plane wave

NOTE The wavelength  $\lambda$  of light in vacuum is given by

$$\lambda = \frac{c}{f}$$

where

$c$  is the speed of light in vacuum  $c = 2,997\,92 \times 10^8$  m/s);

$f$  is the optical frequency

Although the wavelength in dielectric material, such as fibres, is shorter than in vacuum, only the wavelength of light in vacuum is used.

[IEC 60728-6, definition 3.1.17, modified]

**3.1.11****central wavelength**

the average of those wavelengths at which the amplitude of a light source reaches or last falls to half of the maximum amplitude

[IEC 60728-6, definition 3.1.26]

**3.1.12****AM-VSB signal**

sideband in which only the spectral components corresponding to the lower frequencies of the modulating signals are preserved, the other components being strongly attenuated

[IEV 702-06-28, modified]

NOTE This is the abbreviation for the vestigial sideband amplitude modulated signal used in the terrestrial broadcasting and CATV transmission system.

### 3.1.13

#### **QAM signal** **quadrature amplitude modulation** **QAM**

amplitude modulation by two separate signals of two sinusoidal carriers having the same amplitude and frequency but being in phase quadrature, the modulated signals being added for transmission in a single channel

[IEV 702-06-63, modified]

### 3.1.14

#### **OFDM signal**

orthogonal frequency division multiplexing is one of the multiplexing schemes used for the transportation of terrestrial digital broadcasting SDTV and HDTV signals

NOTE OFDM is based on the idea of frequency-division multiplexing, where each frequency channel is modulated with a simpler modulation, and the frequencies and modulation of FDM are arranged to be orthogonal with each other, which almost eliminates the interference between channels.

### 3.1.15

#### **RF signal level definition**

level of an RF signal is defined in Table 1; it is expressed in microvolt or in dB( $\mu$ V) or in dB(mW)

### 3.1.16

#### **AM-VSB analogue signals**

vision carrier signal level is the RMS value of the vision carrier at the peak of the modulation envelope ( $C_{rms}$ ), expressed in dB( $\mu$ V) and measured across a 75  $\Omega$  termination or referred to 75  $\Omega$

NOTE This will correspond, in negative modulation systems, to the carrier amplitude during synchronizing pulses and, in positive modulation systems, to that at peak white level without a chrominance signal, as shown in ITU-R Recommendation BT.470, Figure 1.

### 3.1.17

#### **FM radio or FM audio carrier of a TV signals**

level of an FM radio or of an FM audio carrier of a TV signal is the RMS value of the carrier expressed in dB( $\mu$ V) and measured across a 75  $\Omega$  termination or referred to 75  $\Omega$

### 3.1.18

#### **digitally modulated signals**

level of a digitally modulated signal is given by the RMS power of the signal within the channel bandwidth ( $S_{D,RF}$ ) and can be expressed in dB(mW) or in dB( $\mu$ V) referred to 75  $\Omega$

NOTE The level of an OFDM signal is the average electrical power of the overall signal comprised of multi-carriers and is not the individual carrier level of the multi-carrier signal, as shown in Table 1.

**Table 1 – Level of RF signals**

| Signal             |                      | Level detection | Symbol     | Remarks                                                                                                                           |
|--------------------|----------------------|-----------------|------------|-----------------------------------------------------------------------------------------------------------------------------------|
| Analogue TV signal | AM-VSB video carrier | peak value      | $C_{rms}$  | RMS value of the carrier at the peak of the modulation envelope.                                                                  |
|                    | FM audio carrier     | RMS value       | $C_{rms}$  | The carrier level is a constant value.                                                                                            |
| QAM signal         |                      | RMS value       | $S_{D,RF}$ | The value is averaged over a sufficiently long period of time compared to period of the lowest frequency used for the modulation. |
| OFDM signal        |                      | RMS value       |            |                                                                                                                                   |

**3.1.19****carrier-to-noise ratio** **$C/N$** **signal-to-noise ratio** **$S_{D,RF}/N$** 

ratios are given by

$$\begin{aligned} C/N \text{ (dB)} &= C_{\text{rms}} - N_{\text{rms}} && \text{(for analogue signals)} \\ S_{D,RF}/N \text{ (dB)} &= S_{D,RF} - N_{\text{rms}} && \text{(for digital signals)} \end{aligned}$$

where  $N_{\text{rms}}$  is the RMS level of the noise in the equivalent noise bandwidth of the RF channel, expressed in dB(mW) or in dB(μV) referred to 75 Ω

NOTE The level of the analogue modulated carrier or of the RF digitally modulated signal and the level of the noise shall be expressed in the same units, in dB(mW) or in dB(μV) measured across a 75 Ω termination or referred to 75 Ω.

**3.1.20** **$D/U$  ratio**

ratio of desired signal level,  $D$ [dB(μV)], to undesired signal level,  $U$ [dB(μV)]

NOTE The  $D/U$  ratio is generally used for multiple frequency interference as CSO and CTB, for single frequency interference as CCR.

**3.1.21****single or multiple frequency interference**

besides the  $C/N$  and  $S_{D,RF}/N$  ratios, single or multiple frequency interference to video signal is defined as the ratio of desired signal level and undesired signal level

NOTE 1 The ratio of desired signal level,  $D$ (dB(μV)), to undesired signal level,  $U$ (dB(μV)) is given by

$$D/U \text{ (dB)} = D - U$$

NOTE 2 The desired and the undesired signals can also be expressed both in dB(mW).

**3.1.22****optical line terminal****OLT**

central office-terminal equipment that is linked with the Optical Network Unit (ONU) in customer premises

NOTE OLT usually connects with headend equipment.

**3.1.23****optical network unit****ONU**

terminal equipment linked with OLT

**3.1.24****video-optical network unit****V-ONU**

terminal unit that changes the optical signal of a broadcast system into an electric signal

NOTE The term V-ONU is used as the synonym of optical receiver (O/E) in this standard.

### 3.1.25

#### **stimulated Brillouin scattering**

##### **SBS**

non-linear scattering of optical radiation characterized by a frequency shift as for the Raman scattering, but accompanied by a lower frequency (acoustical) vibration of the medium lattice; the light is scattered backward with respect to the incident radiation

[IEC 61931, definition 2.1.88]

NOTE 1 In silica fibres the frequency shift is typically around 10 GHz.

NOTE 2 SBS results in loss of optical level and affects the performance of analogue optical system.

NOTE 3 The frequency shift is characterized by a frequency downshift (that is to a longer wavelength) due to a GHz frequency acoustical vibration (frequency downshift is 10 or 11 GHz, and gain bandwidth 20 MHz).

### 3.1.26

#### **self-phase modulation**

##### **SPM**

increase of the refractive index of optical fibres due to excessive optical power launched into the fibre

NOTE 1 Modulated optical signal induces modulation of refractive index, which means that the phase of optical signal varies in time. The nonlinear phase modulation is caused by a device or a system itself, therefore it is called self-phase modulation.

NOTE 2 SPM affect the distortion properties of an analogue optical transmission.

### 3.1.27

#### **stimulated Raman scattering**

##### **SRS**

non-linear scattering of optical radiation characterized by a wavelength shift and accompanied by very high frequency vibration of the medium lattice, strongly enhanced by the presence of already scattered radiation

[IEC 61931, definition 2.1.87]

NOTE 1 In silica fibres the wavelength shift is typically around 100 nm for an exciting radiation with a wavelength around 1 550 nm.

NOTE 2 Stimulated Raman scattering can occur in both forward and backward directions and can cause crosstalk between optical signals of different wavelengths.

NOTE 3 Frequency downshift is about 13 THz and gain bandwidth about 20 GHz.

### 3.1.28

#### **cross-phase modulation**

##### **XPM**

caused by the nonlinear refractive index of the fibre material

NOTE 1 XPM has a relationship with the wavelength spacing in an optical transmission system. The more the spacing increases, the more the XPM value decreases. In such a WDM system having 1 490 nm (communication signal) and 1 550 nm (broadcast signal) wavelengths, XPM becomes negligibly small compared to SRS due to this relationship.

NOTE 2 XPM affects the performance of the wavelength division multiplex system.

### 3.1.29

#### **Rayleigh scattering**

light scattering in a medium due to in-homogeneity in material density or composition of that medium which are small with respect to wavelength

NOTE The scattered power is inversely proportional to the fourth power of the wavelength.



[IEV 731-03-37] and [IEC 61931, definition 2.1.76]

### 3.1.30

#### crosstalk

#### carrier-to-crosstalk ratio

#### CCR

level difference of CATV broadcast carrier level and worst case of other services single frequency crosstalk signal measured at RF output port of optical receiver for CATV broadcast service

$$CCR = D_{CATV} - U_{OtherService}$$

where

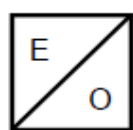
$D_{CATV}$  is the nominal level of CATV broadcast signal in dB(μV) at RF output port of optical CATV broadcast receiver,

$U_{OtherService}$  is the worst case level of another service's single frequency crosstalk in dB(μV) at RF output port of optical CATV broadcast receiver. *The value of  $U_{OtherService}$  is mainly due to the Raman scattering effect.*

CCR is expressed in dB.

### 3.2 Symbols

The following graphical symbols are used in the figures of this standard. These symbols are either listed in IEC 60617 or based on symbols defined in IEC 60617.



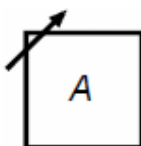
Optical transmitter  
based on  
[IEC 60617-S01231  
(2001-07)]



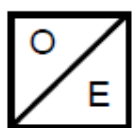
Optical amplifier  
[IEC 60617-S00127  
and  
IEC 60617-S01239  
(2001-07)]



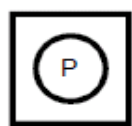
Optical fibre  
[IEC 60617-S01318  
(2001-07)]



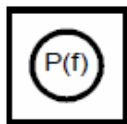
Variable attenuator  
[IEC 60617-S01245,  
modified  
(2001-07)]



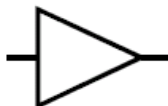
Optical receiver  
based on  
[IEC 60617-S00213  
(2001-07)]



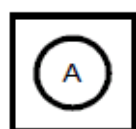
Power meter  
[IEC 60617-S00059,  
IEC 60617-S00910  
(2001-07)]



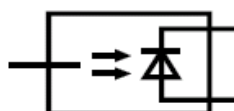
Electrical spectrum  
analyzer  
based on  
[IEC 60617-S00059  
and  
IEC 60617-S00910  
(2001-07)]



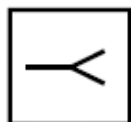
Amplifier  
[IEC 60617-S01239  
modified  
(2001-07)]



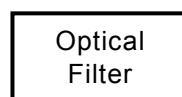
Ammeter  
based on  
[IEC 60617-S00059  
and  
IEC 60617-S00910  
(2001-07)]



Photodiode with fibre  
pigtail  
[IEC 60617-S01327  
(2001-07)]



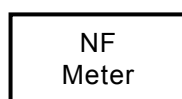
Coupler  
[IEC 60617-S00059  
and  
IEC 60617-S01188  
(2001-07)]



Optical filter



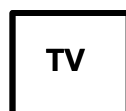
Optical terminator  
[IEC 60617-S01389  
and  
IEC 60617-S01318<sup>2</sup>]



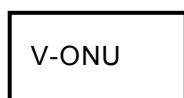
NF meter



Optical splitter  
[3.33.1 of IEC 61930]



Television set



Video optical network  
unit

### 3.3 Abbreviations

|       |                                                                        |       |                                                                  |
|-------|------------------------------------------------------------------------|-------|------------------------------------------------------------------|
| ADS   | Active Double Star                                                     | AGC   | Automatic Gain Control                                           |
| AM    | Amplitude Modulation                                                   | APC   | Angled Physical Contact optical connector                        |
| BCH   | Bose-Chaudhuri-Hocquenghem multiple error correction binary block code | CATV  | Community Antenna Television (network)                           |
| CODFM | Coded Orthogonal Frequency Division Multiplex                          | CCR   | Carrier-To-Crosstalk ratio                                       |
| C/N   | Carrier-to-Noise ratio                                                 | CPE   | Customer Premises Equipment                                      |
| CSO   | Composite Second Order                                                 | CTB   | Composite Triple Beat                                            |
| DS    | Down Stream or Double Star                                             | CW    | Continuous Wave                                                  |
| D/I   | Desired to Undesired signal ratio                                      | DSF   | Dispersion Shifted Fibre                                         |
| E/O   | Optical transmitter (Electrical-to-Optical transducer)                 | EDFA  | Erbium-Doped Fibre Amplifier                                     |
| FTTB  | Fibre To The Building                                                  | FM    | Frequency Modulation                                             |
| HDTV  | High Definition Television                                             | FTTH  | Fibre To The Home                                                |
| HFC   | Hybrid Fibre Coaxial                                                   | H/E   | Headend                                                          |
| LD    | Laser Diode                                                            | ITU-T | International Telecommunication Union – Telecommunication sector |
|       |                                                                        | LDPC  | Low-Density Parity Check (codes)                                 |

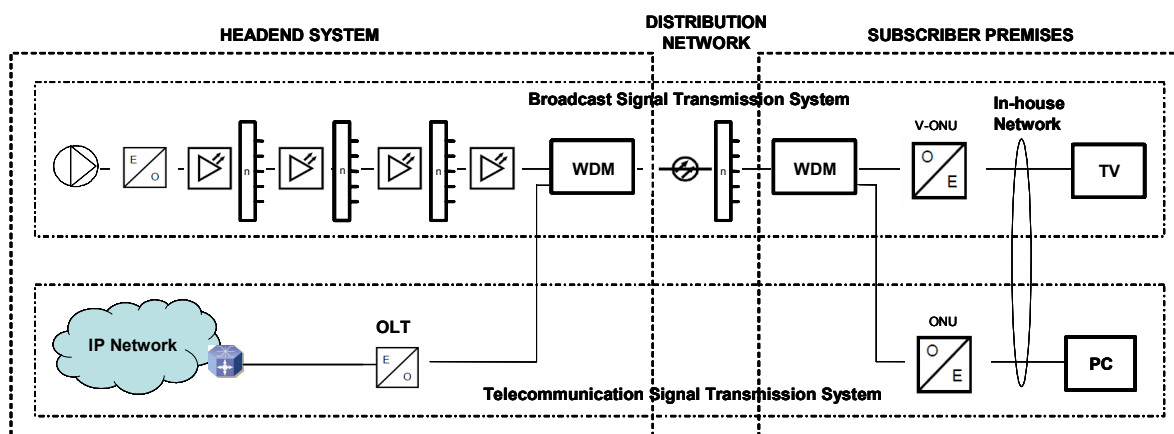
<sup>2</sup> IEC 60617-S01318 is for reference only, obsolete since 1996-05.

|       |                                                     |        |                                          |
|-------|-----------------------------------------------------|--------|------------------------------------------|
| MC    | Media Converter                                     | MDU    | Multiple Dwelling Unit                   |
| MER   | Modulation Error Ratio                              | NF     | Noise Figure                             |
| O/E   | Optical Receiver (Optical to Electrical transducer) | OFCS   | Optical Fibre Communication System       |
| OFDM  | Orthogonal Frequency Division Multiplex             | OLT    | Optical Line Terminal                    |
| OMI   | Optical Modulation Index                            | ONU    | Optical Network Unit                     |
| PD    | Photo Diode                                         | PDS    | Passive Double Star                      |
| PER   | Packet Error Ratio                                  | PON    | Passive Optical Network                  |
| QAM   | Quadrature Amplitude Modulation                     | QPSK   | Quaternary Phase Shift Keying            |
| RIN   | Relative Intensity Noise                            | RBW    | Resolution Bandwidth                     |
| RF    | Radio Frequency                                     | SBS    | Stimulated Brillouin Scattering          |
| SDTV  | Standard Definition Television                      | SDU    | Single Dwelling Unit                     |
| SMF   | Single Mode Fibre                                   | S/N    | Signal-to-Noise ratio                    |
|       |                                                     | SPM    | Self-Phase Modulation                    |
| SRS   | Stimulated Raman Scattering                         | SS     | Single Star                              |
| US    | Up Stream                                           | VBW    | Video Bandwidth                          |
| V-ONU | Video Optical Network Unit                          | AM-VSB | Amplitude Modulation-Vestigial Side Band |
| WDM   | Wavelength Division Multiplexing                    | XPM    | Cross-Phase Modulation                   |
| SW    | Switch                                              |        |                                          |

#### 4 Optical system reference model

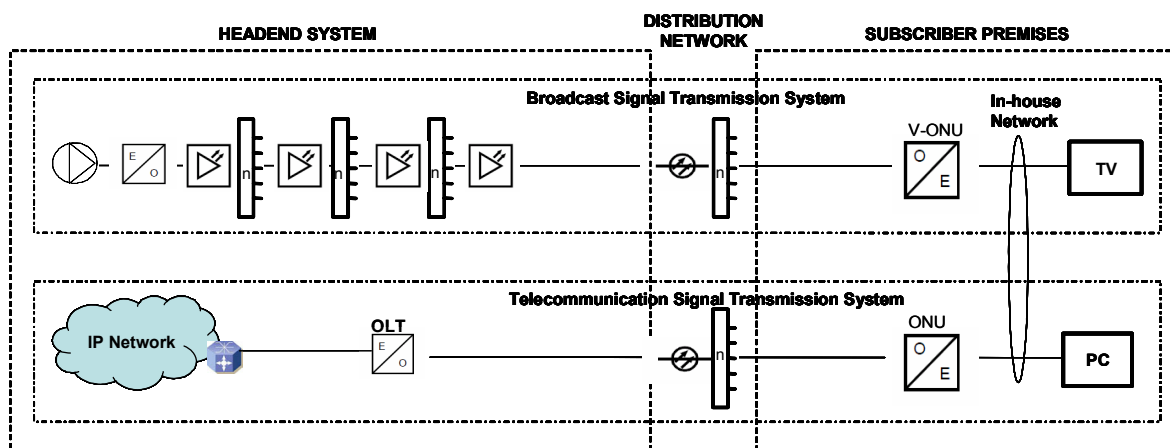
Figure 1 and Figure 2 show the optical system reference model for broadcast signal transmission. Although numbers of optical amplifiers and optical splitters depend on the scale of the optical system, or on the numbers of subscribers to be connected, the fundamental network configuration shall follow the system reference model. In addition the optical levels in operation required for the system are relatively high, and hence, special attention for human safety shall be paid in accordance with IEC 60825-1, IEC 60825-2 and IEC 60825-12.

Generally, there are two solutions for constructing an optical transmission system: one-fibre and two-fibre solutions. One-fibre solution is suitable for deployment of both broadcast and telecommunication services cost-effectively using Wave Division Multiplex (WDM) technology. However, special attention shall be paid to the selection of the transmission parameters to avoid mutual interference over single optical fibre transmission. The two-fibres solution is suitable for segregation of service areas for broadcast and telecommunications and is free from mutual signal interference. However, the installation cost is not competitive compared with the one-fibre solution.



IEC 2539/09

Figure 1 – Optical system reference model for one-fibre solution



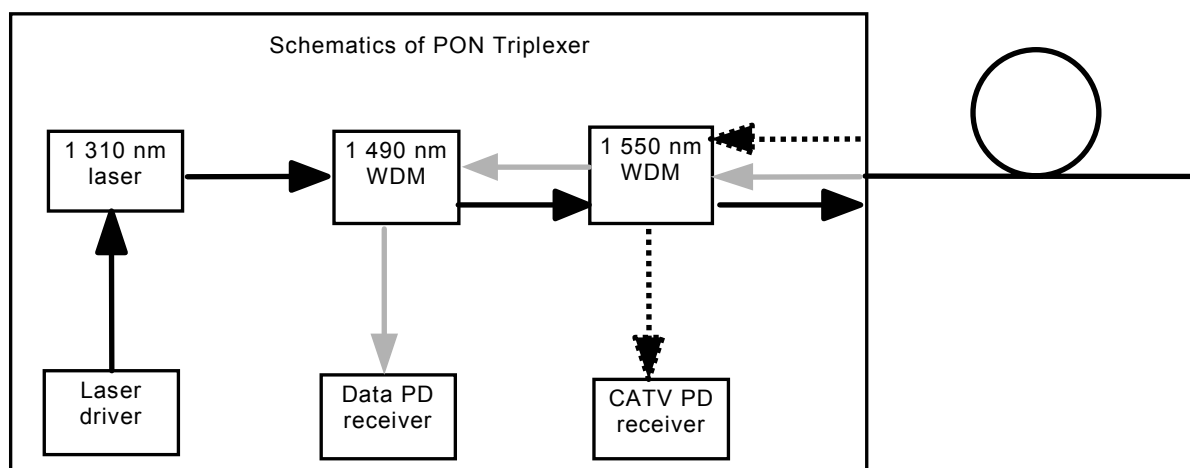
IEC 2540/09

Figure 2 – Optical system reference model for two-fibres solution

In this standard the one-fibre solution is used as the reference model.

The reference model as shown in Figure 1 includes the broadcast signal transmission system and the telecommunication signal transmission system. A telecommunication signal transmission system uses both ways of transmission over the optical fibre with different optical wavelengths. Both systems are combined by WDM filters at input and output of the distribution network as an example. The distribution network must be passive optical components such as optical fibre and optical power splitters, considering maintenance and future system expansion. Although the telecommunication signal transmission system can transmit any IP datagram to subscriber premises through IP networks, this system is out of the scope of this standard.

In some cases a one-fibre solution triplexer is used. The schematics of the PON triplexer is given in Figure 3.



IEC 2541/09

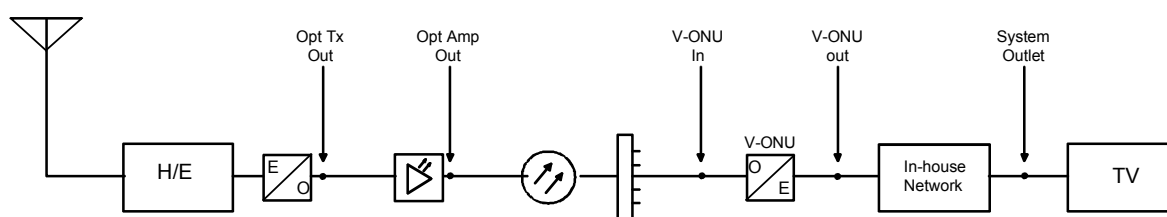
**Figure 3 – Example of PON triplexer**

In the one-fibre solution it is required to avoid optical and electrical crosstalk.

- Optical crosstalk between the downstream data signal transmitted on 1 490 nm wavelength and the CATV downstream signal is generated if the 1 550 nm WDM does not provide sufficient isolation for the 1 490 nm signal.
- Electrical crosstalk between the 1 310 nm data driver signal and the 1 550 nm CATV PD receiver input signal due to electromagnetic radiation in the very compact triplexer housing. Example: Typical laser driver currents of the 1 310 nm laser are in the order of 10 dB(mA). The input receiver current of one TV signal is in the order of -36 dB(mA), that means 46 dB lower. In order to have less than -60 dB crosstalk between the IP transmitter and the CATV receiver signal, the isolation of the IP driver signal respect to the CATV receiver has to be greater than 100 dB. This is quite difficult to achieve in compact housings for a wide frequency range from about 45 MHz to 1 000 MHz.

Therefore in some systems, PON diplexers with separate 1 550 nm WDMs are used instead of PON triplexers to achieve better performance for CATV transmission.

The performance specified points of the optical system are described in Figure 4. Some of the performance check point in the optical distribution network should be provided in accordance with maintenance policy of cable operator.



• Performance Specified Point

IEC 2542/09

**Figure 4 – Performance specified points of the optical system**

## 5 Preparation of measurement

### 5.1 Environmental conditions

#### 5.1.1 Standard measurement conditions

Unless otherwise specified, all the measurements shall be carried out under the following standard measurement conditions.

### 5.1.2 Temperature and humidity

The ambient temperature and relative humidity shall be in the range of 15 °C to 35 °C and 25 % to 75 %, respectively, see IEC 60068-1, 5.3.1] nevertheless the specification of the measurement equipment has to be taken into account.

### 5.1.3 Setting up the measuring setup and system under test

The system under test shall be in the normal operating condition, and all the pieces of equipment in the system shall be mounted and tuned according to the designed level diagram prior to the measurement.

### 5.1.4 AGC operation

Unless otherwise specified, all the pieces of equipment in the system shall be operated in the AGC mode if available.

### 5.1.5 Impedance matching between pieces of equipment

Attention shall be paid on the impedance matching between pieces of equipment and the test setup, and sufficient care shall be taken to avoid any measurement error by introducing components such as attenuators.

### 5.1.6 Standard operating condition

The standard operating condition refers to the condition in which the cable TV system under test is fully functional at a given facility. All the input and output of individual pieces of equipment shall be tuned according to the designed level diagram before any measurement is carried out.

### 5.1.7 Standard signal and measuring equipment

For measurement purposes, the standard signals used in the measuring instruments as well as in the system under test shall be set according to the prescribed standard signal format of the individual system. The measuring instruments to be used are described in Table 2 (passive pieces of equipment are excluded).

**Table 2 – Measuring instruments**

| Name of instrument                                                                                                                                                                                                                                        | Usage                                                                                  |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------|
| Optical power meter                                                                                                                                                                                                                                       | Instrument to measure the power of the optical signal.                                 |
| Spectrum analyzer                                                                                                                                                                                                                                         | Instrument used for quantitative measurement of high frequency signals.                |
| Signal generator                                                                                                                                                                                                                                          | Instrument used to generate high frequency signals (sine-waves).                       |
| Network analyzer <sup>a</sup>                                                                                                                                                                                                                             | Instrument used to measure the high frequency performance of equipments.               |
| NF meter <sup>a</sup>                                                                                                                                                                                                                                     | Instrument used to measure Noise Figure ( <i>NF</i> ).                                 |
| Current meter (Ammeter) <sup>a</sup>                                                                                                                                                                                                                      | Instrument used to measure electrical current.                                         |
| V-ONU                                                                                                                                                                                                                                                     | Optical receiver unit used to convert an optical video signal to an electrical signal. |
| <sup>a</sup> If the <i>RIN</i> calculation parameters of ONU, responsivity ( <i>R</i> ), dark current ( <i>I<sub>do</sub></i> ) and equivalent noise current density ( <i>I<sub>eq</sub></i> ) are known beforehand, these instruments are not necessary. |                                                                                        |

## 5.2 Accuracy of measuring equipment

All the devices and instruments used for the measurement shall be accurately calibrated. The standard sources used for calibration shall be calibrated within 6 months before the day of measurement.

## 5.3 Source power

The supply voltage and frequency for the measuring instruments and the equipment of the system under test shall be obtained from the corresponding instrument/equipment specifications.

# 6 Methods of measurement

## 6.1 Measuring points and items

### 6.1.1 General

This clause describes methods of measurement specifically designed for FTTH system.

The measurement points described in this standard are limited to the part of the system, that is ranging from the output terminal of the optical transmitter to the system outlet.

### 6.1.2 Measuring points

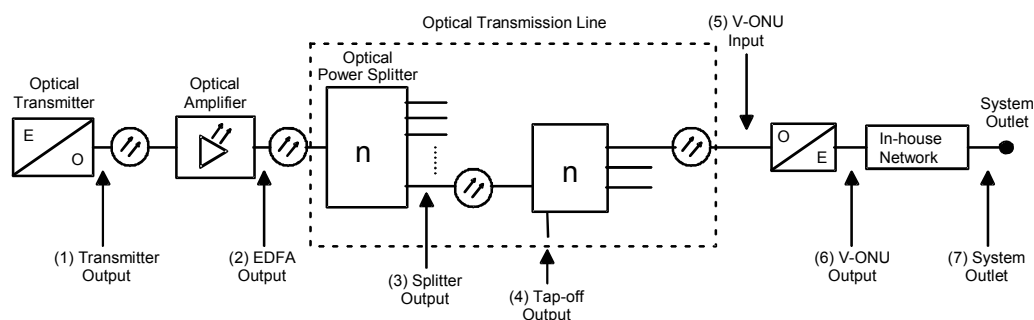
It is required to measure the optical power at points (1) to (5), and the electrical signal level at points (6) and (7) of Figure 5 to assure the total system performance. Points (5), (6) and (7) shall be measured to guarantee the system performance at the end point of the optical section and at the interface point to the customer premises. *R/N* should be measured at points (1) to (5) and *C/N* (electrical signal) at points (6) and (7).

### 6.1.3 Measured parameters

The following measurements have to be carried out.

#### a) Optical power

The optical power shall be measured at points (1) to (5) in the optical system shown in Figure 5.



IEC 2543/09

**Figure 5 – Typical optical video distribution system**

#### b) Carrier-to-noise ratio (electrical signal)

The carrier-to-noise ratio is measured after the optical signal is converted into an electrical signal, and it shall be carried out at measurement points (6) and (7).

c) *C/N* ratio (*RIN*)

Estimation of the carrier-to-noise ratio at the output of V-ONU is calculated from the measured *RIN* (relative intensity noise) of the optical input signal of V-ONU at point (5).

It is preferable to measure the *RIN* when the optical power at the measuring point is higher than –3 dB(mW), a limitation imposed by the noise performance of the measuring setup. Similarly, since the optical power at the measuring point (5) in a typical system is lower than –3 dB(mW), the measurement error becomes large and the measurement of *RIN* at this point is not recommended.

However, since the above limitation is due only to the noise performance of the measuring system, this can be exempted if the accuracy of measurement improves in future.

For measuring points and measured parameters see Table 3.

**Table 3 – Measuring points and measured parameters**

| Measured parameters                                                                                                                                | Measuring points          |                    |                              |                       |                    |                     |                      |
|----------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------|--------------------|------------------------------|-----------------------|--------------------|---------------------|----------------------|
|                                                                                                                                                    | (1)<br>Transmitter output | (2)<br>EDFA output | (3)<br>Power splitter output | (4)<br>Tap-off output | (5)<br>V-ONU input | (6)<br>V-ONU output | (7)<br>System outlet |
| Optical power                                                                                                                                      | ○                         | ○                  | ○                            | ○                     | ○                  | —                   | —                    |
| <i>C/N</i> (electrical)                                                                                                                            | —                         | —                  | —                            | —                     | —                  | ○                   | ○                    |
| <i>C/N (RIN)</i><br>(see NOTE 1)                                                                                                                   | ○                         | ○                  | △                            | △                     | △                  | —                   | —                    |
| ○ : Measurements are possible at these points.<br>△ : Measurements are possible at these points when the optical power is higher than –3 dB (mW).  |                           |                    |                              |                       |                    |                     |                      |
| NOTE 1 Theoretical estimation of <i>C/N</i> at (6), at the output of V-ONU, is based on the measurement results of individual pieces of equipment. |                           |                    |                              |                       |                    |                     |                      |
| NOTE 2 The measurement at points (5), (6) and (7) is mandatory, while measurement at other points is required to assure the system performance.    |                           |                    |                              |                       |                    |                     |                      |

## 6.2 Optical power

### 6.2.1 General

The purpose of this measurement is to measure the average optical power at each measurement point in the optical system illustrated in Figure 6 or in Figure 7.

### 6.2.2 Measuring setup

#### 6.2.2.1 Measurement of the optical power at single wavelength

The measurement of optical power at single wavelength shall be carried out according to 4.2 of IEC 60728-6.

#### 6.2.2.2 Measurement of the optical power of a WDM signal

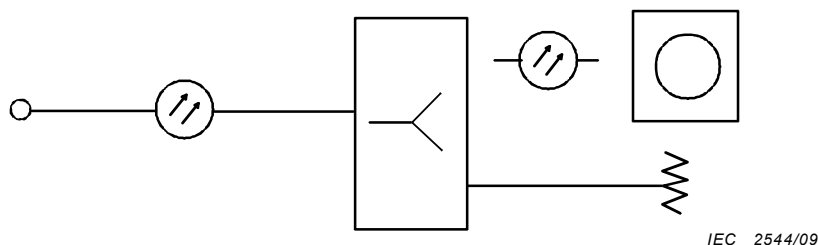
When multiple wavelengths are multiplexed, either by using an optical filter or a WDM coupler, the optical power of the specified wavelength shall be measured. The directivity and isolation performance of the WDM coupler used for the measurement shall be the same or better than



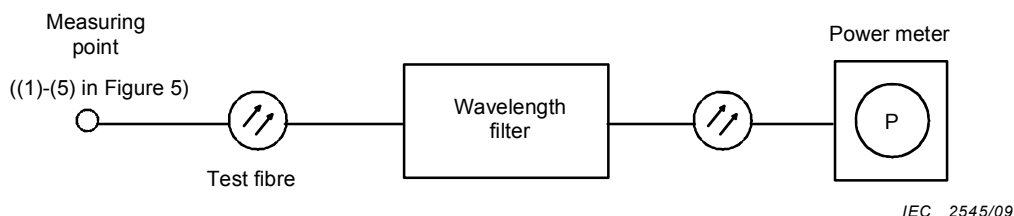
the filter used inside the V-ONU. Connect the equipments as shown in Figure 6 or in Figure 7 depending on whether a WDM coupler or a wavelength filter is used.

NOTE Methods of measurement for the optical power of single wavelength are described in IEC 60728-6.

The optical power at the output of a WDM coupler shall be measured according to 4.2 of IEC 60728-6.



**Figure 6 – Measurement of optical power using a WDM coupler**



**Figure 7 – Measurement of optical power using a wavelength filter**

### 6.2.3 Measuring method

In addition to the requirements of IEC 60728-6, the following points shall be considered.

- Connect the test fibre to the power meter and ensure that the power meter is switched to the specified single wavelength and only the specified wavelength signal is applied to the power meter.
- Measure and record the power of the output signal using the power meter.
- If a WDM coupler or a wavelength filter is used to measure the WDM signals, the pre-determined insertion loss of the WDM coupler or the wavelength filter shall be added to the measured optical power.

### 6.2.4 Precaution for measurement

The following considerations have to be taken into account.

- Optical fibre end-face or the connector end-face should not be viewed directly. Also, the end-face of the fibre should not be pointed towards any person. If there are unterminated single or multiple fibres, they shall be covered together to avoid any radiation hazard. Any unconnected optical connector shall be covered with a cap during all the measuring time.
- Ensure that the power meter has a measuring range suitable for the expected power, and is capable of measuring the expected signal wavelength. The detector system of the power meter shall have a sufficiently large area to collect all the radiation from the test fibre and a spectral sensitivity compatible with the light source. A minimum accuracy of  $\pm 10\%$  is recommended.
- The sensor portion of the power meter shall be shut off and zero offset adjustment shall be carried out before the measurement.
- Test fibres and connectors shall have clean and unscratched ends in order to prevent losses of power and reflections.

- e) If the measurement bandwidth and the measuring range of the power meter can be set independently, they shall be set in the auto mode prior to the measurement.
- f) The measurement shall be carried out in the CW mode. If the power meter has a selectable measurement mode (CW / 270 Hz / 1 kHz / 2 kHz), CW mode shall be selected. Power meters without any measurement mode function, normally operate with in CW mode and ensure CW mode of operation prior to the measurement.
- g) If there is any instability in the measurement (measured value may fluctuate when end-face of fixed attenuator is directly connected to the power meter), a suitable patch cord shall be used and the measurement shall be repeated.
- h) The potential sources of error are the following:
  - 1 the inaccuracy of the power meter, for example if its dark current is not sufficiently low;
  - 2 the attenuation of the test fibre and the specified coupling means.

## 6.2.5 Presentation of the results

The optical power shall be expressed in dB(mW).

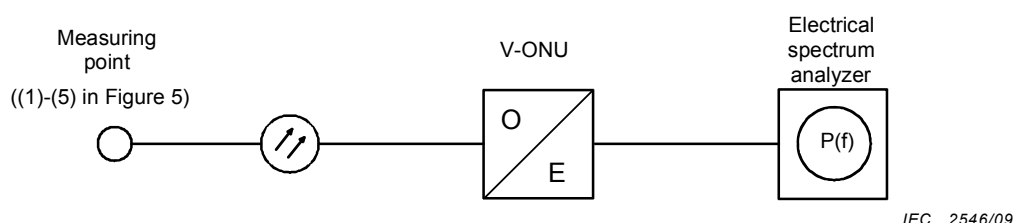
## 6.3 Carrier level and carrier-to-noise ratio

### 6.3.1 General

The purpose of this test method is to measure the carrier level of the television broadcast signal. Also, carrier-to-noise ratio is measured using the measured noise level within the transmission bandwidth of the television signal. This test method performs the measurement in the electrical domain.

### 6.3.2 Measuring setup

Connect the pieces of equipment as shown in Figure 8. The method for measuring the carrier-to-noise ratio of analogue optical transmission systems is nearly the same as for cabled distribution systems (see IEC 60728-1, 4.6)



**Figure 8 – Arrangement of test equipment for carrier-to-noise ratio measurement**

### 6.3.3 Measuring conditions

The following measuring conditions apply.

- a) The spectrum analyzer used for the measurement has to be calibrated before the measurement. The supply voltage of all the pieces of equipment used for the measurements shall be switched on at least 30 min before the start of the measurement.
- b) If the measuring instrument has any calibration function, it shall be executed prior to the measurement.
- c) Suitable coaxial cables and connectors shall be used to maintain proper impedance matching within the measurement system.

### 6.3.4 Measuring method for analogue signals (AM-VSB)

Methods of measurement for the carrier level and carrier-to-noise ratio ( $C/N$ ) are described in 4.6 and 4.8 of IEC 60728-1.

### 6.3.5 Measuring method for digitally modulated signals (64 QAM, OFDM)

Methods of measurement for the carrier level and carrier-to-noise ratio for digitally modulated signals are described in IEC 60728-1.

### 6.3.6 Precautions for measurement

The following considerations have to be taken into account.

- a) Measurement accuracy: To obtain accurate measurement of carrier-to-noise ratio, it is necessary to turn off the channel under test and measure the noise level within the channel bandwidth. Depending on the situation, it is expected that the arbitrary broadcast channel cannot be turned off during network operation. Therefore, attention shall be paid on the inaccuracy and measuring error of the test method prescribed in this standard.
- b) Attenuation setup of the spectrum analyzer: Most of the spectrum analyzers have a default input attenuation of 10 dB when powered on. It is possible to carry out the measurement with this default value when the total electrical input power does not exceed 0 dB(mW). Total electrical power is measured using the electrical power measurement option of the spectrum analyzer and by setting the centre frequency to 510 MHz and the frequency span to 1 GHz and the channel power measurement bandwidth to 1 GHz. There shall not be any signal outside the above frequency span. If the total power ( $P_T$ ) exceeds 0 dB(mW) (109 dB( $\mu$ V) on the voltage display), in order to avoid any distortion generated within the spectrum analyzer, adjust the input attenuation ( $ATT_{IN}$ ) setting to satisfy the following relation:

$$P_T - ATT_{IN} < -10 \text{ dB(mW)}.$$

- c) Measure the output of the digital signal generator using a calibrated power meter with a thermo-coupled sensor and, taking this as the true value, calibrate the measurement level of the spectrum analyzer.
- d) When the carrier-to-noise ratio of the signal is very small, the noise within the signal will be larger than the measuring error and cannot be neglected. If a correction factor needs to be applied, this correction factor shall be subtracted from the spectrum analyzer measured level (see Annex F of IEC 60728-1).
- e) This standard recommends the test method using the electrical power measurement option of the spectrum analyzer to deal with QAM and OFDM signals. This test method is preferred because any correction necessary within the spectrum analyzer is automatically processed irrespective of the type of spectrum analyzer used for the measurement. Also, the flatness of the signal over the transmission bandwidth does not influence the measurement results.

NOTE The noise level per unit frequency may be expressed in dB( $\mu$ V/ $\sqrt{\text{Hz}}$ ) or in dB( $\mu$ V/Hz).

### 6.3.7 Presentation of the results

The carrier level shall be expressed in dB(mW) or in dB( $\mu$ V) and the carrier-to-noise ratio shall be expressed in dB.

## 6.4 Carrier-to-noise ratio defined by optical signal

### 6.4.1 General

This measurement method has the purpose to predict the carrier-to-noise ratio at the output of V-ONU from the measured relative intensity noise ( $RIN$ ) of the optical input signal to the V-ONU.

$RIN$  is the noise caused by fluctuations in optical output power with respect to time and is expressed as the ratio of average optical power to the average noise power measured in 1 Hz bandwidth. It is difficult to measure the  $RIN$  directly in the optical domain and the measurement shall be carried out after converting the optical signal to an electrical signal. However, an accurate measurement of  $RIN$  is not possible if the optical input to V-ONU is

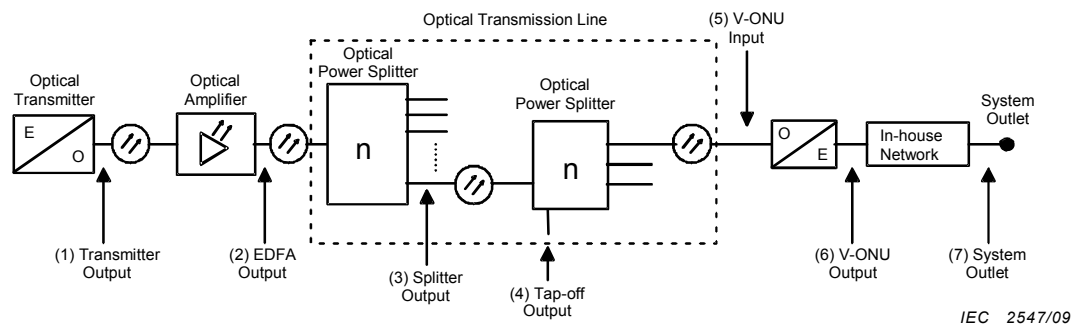
small as in most of the practical systems.  $R/N$  may also be calculated from the measured performance of individual components constituting the system. However, it is necessary to measure the  $R/N$  on a near side measurement point.

#### 6.4.2 Measuring setup

The measuring setup is the following.

##### a) Measuring points

Measuring points in the Cable TV network for optical signals are shown in Figure 9.



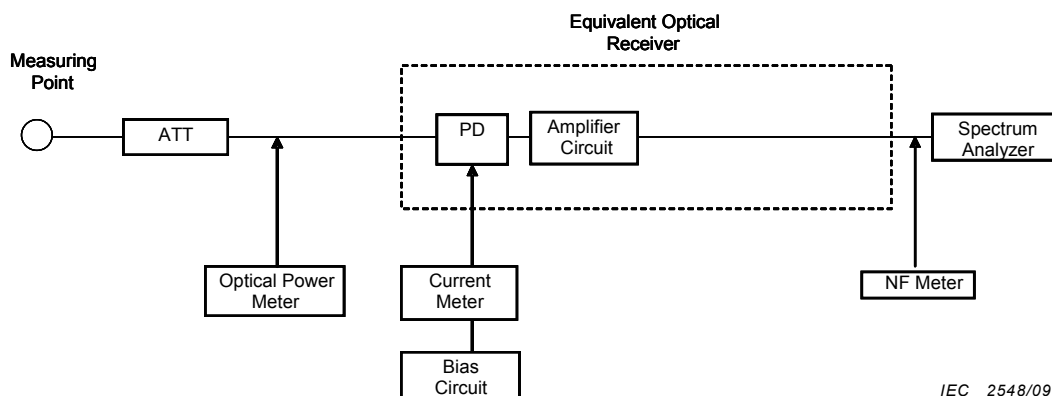
NOTE Figure 9 is identical to Figure 5, except that the Figure title has changed to describe the measurement points.

**Figure 9 – Measuring points in the optical cable TV network**

- In order to calculate the carrier-to-noise ratio at the V-ONU output, it is necessary to measure the  $R/N$ , as shown in Figure 9 at points (1) to (3), where the optical output power is sufficiently high to allow  $R/N$  measurements to be accurate.
- NOTE  $R/N$  measurements will not be accurate when the optical power is lower than  $-3$  dB(mW).
- If an optical amplifier is not employed in the system,  $R/N$  shall be measured at point (1).
- If an outdoor type optical amplifier is employed and measurement can be carried out outdoor, the optical amplifier output shall be considered as a measuring point.
- If the optical power at point (4) or (5) is sufficiently high, these points shall also be used for measuring  $R/N$ .

##### b) Measuring setup

Figure 10 shows the  $R/N$  measurement set-up.



**Figure 10 – *RIN* measurement setup**

### 6.4.3 Measuring conditions

The following measuring conditions apply.

- Only calibrated instruments (spectrum analyzer, optical power meter, current meter, network analyzer, *NF* meter and the optical attenuator) shall be used for the measurements.
- The spectrum analyzer must have the option to measure the noise power density. The optical receiver part is constituted by a photo diode (PD), a low-noise preamplifier and a matching circuit. The photo diode must have the provision to measure the photo diode current.
- A CW optical signal shall be used for the measurement. To avoid the SBS interference some technology shall be applied such as SBS suppression carrier method.
- The optical input level to the optical receiver shall be around 0 dB(mW), and shall not be lower than –3 dB(mW).

The *RIN* degradation due the Rayleigh scattering and multiple optical reflections within the transmission line cannot be neglected. Therefore, if the *RIN* measurement is carried out within the head-end, an equivalent optical cable having similar performance to the cable used in the actual optical network, shall be inserted at the measuring point in Figure 10.

### 6.4.4 System *RIN* measuring method

#### 6.4.4.1 General

This test method shall be applied to predict the carrier-to-noise ratio at the output of V-ONU from the *RIN* measurement using the setup shown in Figure 10. This subclause contains several steps as shown below. If the parameters for  $R$ ,  $I_{d0}$ ,  $I_{eq}$  and  $G$  are unknown, refer to Annex D. *RIN* can be calculated using these parameters.

#### 6.4.4.2 STEP A: Input power of optical receiver and system noise (noise current density)

For step A proceed as follows.

- Measure the input power of optical receiver ( $P_r$ ) using a power meter.
- Connect the spectrum analyzer at the output of the optical receiver and select the measurement mode to measure the noise power density. Measure the noise power density per unit frequency,  $N_p$  expressed in dB(mW/Hz). The total noise current per Hz,  $I_{bn}$  of the optical receiver can be calculated using Equation (1) with RBW of the spectrum analyzer set to 100 kHz)

$$I_{bn} = \sqrt{\frac{10^{\frac{N_p}{10}} \times 10^{-3}}{Z_0}} \quad [A/\sqrt{Hz}] \quad (1)$$

where

$Z_0$ : is the impedance of the measurement setup,  
 $N_p$  is the noise power density, expressed in dB(mW/Hz).

The following correction shall be applied if the noise level ( $N_L$ ) is measured with the spectrum analyzer:

$$N_p = N_L + 10 \log (B_n/B) + K_1 + K_2$$

where

$B_n$  is the measurement bandwidth of noise power ( $N_p$ ) 1 Hz,  
 $B$  is the noise bandwidth,  $RBW \times 1,2$ (noise bandwidth correction factor) = 120 000 Hz,  
 $K_1$  is the correction factor for conversion to effective voltage level =  $10 \log(2/\sqrt{\pi}) = 1,05$  dB,  
 $K_2$  is the correction factor for the logarithmic amplifier of spectrum analyzer = 1,45 dB.

NOTE The measured noise level ( $N_p$ ) includes that of the measuring equipment (spectrum analyzer) which should be at least 20 dB lower than the noise level displayed outside the channel band in order not to affect the results. Otherwise, the contribution of noise (due to the system or the equipment under test and to the measuring equipment) should be taken into account in the measurement of noise level (see Annex F of IEC 60728-1).

#### 6.4.4.3 STEP B: $RIN$ calculation

For step B proceed as follows.

- From the above measurement results,  $RIN$  can be calculated from the following relation:

$$RIN = 10 \lg \left( \frac{\frac{I_{bn}^2}{G}}{(R \times P_r)^2} - \frac{2e}{(R \times P_r)^2} (I_{d0} + R \times P_r) - \frac{I_{eq}^2}{(R \times P_r)^2} \right) [dB(Hz^{-1})] \quad (2)$$

where

$R$  is the responsivity of the photodiode (A/W),  
 $I_{d0}$  is the dark current of the photodiode (A),  
 $I_{eq}$  is the preamplifier equivalent input noise current density (A/ $\sqrt{Hz}$ ),  
 $I_{bn}$  is the total noise current within 1 Hz bandwidth at the optical receiver output (A/ $\sqrt{Hz}$ ),  
 $G$  is the amplifier gain of the optical receiver (Including gain of matching circuit)  
 $P_r$  is the input power to the optical receiver (W),  
 $e$  is the charge of the electron  $1,602 \times 10^{-19}$ (C).

#### 6.4.5 C/N calculation based on *RIN* value

The carrier-to-noise ratio (C/N) at the V-ONU output can be calculated using the following relation.

$$C/N = 10 \lg \left( \frac{1}{B_N} \cdot \frac{\frac{1}{2} \cdot (m_k \cdot R \cdot P_r)^2}{RIN(R \cdot P_r)^2 + 2 \cdot e \cdot (I_{d0} + R \cdot P_r) + I_{eq}^2} \right) \text{ [dB]} \quad (3)$$

where

$$M = \sqrt{\sum_{k=1}^K m_k^2} \quad (4)$$

The other parameters for the calculation are listed in Table 4.

#### 6.4.6 Component *RIN* calculation

The following method shall be applied to calculate the component *RIN* of the optical signal at input of the V-ONU when 6.4.4 is not applicable. If the *RIN* of the first EDFA (*RIN* of optical transmitter) is expressed as *RIN*<sub>in</sub>, then the *RIN* of the *n*th EDFA, *RIN*<sub>out</sub> is given by

$$RIN_{out} = 10 \lg \left( \sum \frac{2 \cdot E \cdot 10^{\frac{NF_n}{10}}}{10^{\frac{P_n}{10}}} + 10^{\frac{RIN_{in}}{10}} \right) \quad (5)$$

where

- $E$  is the photon energy,  $E = hf$ ,  
 $h$  is the Planck's constant,  $6,62 \times 10^{-34}$ [Js],  
 $f$  is the frequency.  
 If the optical wavelength is 1 555 nm, then  $E = 1,278 \times 10^{-16}$ [mJ].
- $NF_n$  is the noise factor of the *n*th EDFA (dB),
- $P_n$  is the optical input power of the *n*th EDFA (dB(mW)).

NOTE "1/G" term in Equation (12) of IEC/TR 60728-6-1 is very small compared to other terms and hence can be neglected.

- Also, even though the *RIN* degradation due to Rayleigh scattering and other reflections within the fibre is small, this cannot be ignored if an optical transmitter with *RIN* smaller than –160 dB(Hz<sup>-1</sup>) and EDFAs with low *NF* are used. The following relation shall be used to calculate the *RIN*, *RIN*<sub>f</sub> due to the fibre transmission.

$$RIN_f = 10 \lg \left[ \frac{s^2}{4} (2\alpha L - 1 + e^{-2\alpha L}) \cdot \frac{\Delta \nu}{\pi(f_{RF}^2 + \Delta \nu^2)} \right] \text{ (dB/Hz)} \quad (6)$$

where

- $S$  is the ratio of scattered optical power that is propagated in the reverse direction,  
 $s = \frac{1,5}{(\pi \cdot W \cdot \eta_1)^2}$
- $\alpha$  is the fibre transmission loss. If the transmission loss is  $\alpha_{dB}$ (dB/km), then  $\alpha = \alpha_{dB}/4,343$ .

- $L$  is the transmission distance (km),
- $\Delta\nu$  is the spectral width of the optical signal when modulated (Hz),
- $f_{RF}$  is the measurement frequency (Hz),
- $W$  is the fibre mode field diameter ( $\mu\text{m}$ ),
- $\eta_i$  is the refractive index of fibre core.

- The  $R/N$  of the optical signal at the input of V-ONU is given by

$$R/N = -10 \lg \left[ 10^{-(R/N_{\text{out}}/10)} + 10^{-(R/N_f/10)} \right] \quad (\text{dB/Hz}) \quad (7)$$

Based on the  $R/N$  value above,  $C/N$  can be calculated by Equation (3).

**Table 4 – Parameters used for the calculation of carrier-to-noise ratio ( $C/N$ )**

| Parameter | Remarks                                                                                                                                                                                       |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
|-----------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| $B_N$     | Noise bandwidth<br>AM-VSB: 4,00 MHz (NTSC)<br>5,08 MHz (I)<br>4,75 MHz (B, G, D1)<br>5,00 MHz (L)<br>5,75 MHz (D, K)<br>QAM: Table H.1 of Part-1 Annex H<br>OFDM: Table H.1 of Part-1 Annex H | This parameter depends on transmission signal format.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
| $K$       | Number of transmission carriers                                                                                                                                                               |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
| $M$       | Total optical modulation index                                                                                                                                                                |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
| $m_k$     | Optical modulation index of $k^{\text{th}}$ carrier<br>(modulated RF carriers)                                                                                                                | These parameters depend on optical transmitter, transmission signal, etc.                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
| $P_r$     | Received optical power (W)                                                                                                                                                                    | This parameter depends on transmission line design.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
| $R/N$     | $R/N$ of the optical signal input to the V-ONU ( $\text{dB}(\text{Hz}^{-1})$ )                                                                                                                | This parameter depends on optical transmitter, amplifier and transmission line. If the parameter is unknown, the following values may be used to calculate the $R/N$ of optical signal input to the V-ONU.<br><br>$R/N$ of optical transmitter for multi-channel transmission is $-155 \text{ dB}(\text{Hz}^{-1})$ .<br><br>$R/N$ of optical transmitter for retransmission is $-150 \text{ dB}(\text{Hz}^{-1})$ .<br><br>$NF$ of optical amplifier is 6,5 dB<br><br>$R/N$ due to optical transmission line is $-161 \text{ dB}(\text{Hz}^{-1})$ . |
| $e$       | Charge of an electron ( $1,602 \times 10^{-19} \text{ C}$ )                                                                                                                                   | Physical constant.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
| $R$       | Responsivity of V-ONU (A/W)                                                                                                                                                                   | ...                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
| $I_{d0}$  | Dark current of V-ONU (A)                                                                                                                                                                     |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |

### Example for calculating carrier-to-noise ratio ( $C/N$ )

Carrier-to-noise-ratio ( $C/N$ ) may be calculated as follows with the following typical parameters:



|                                                                      |                                        |
|----------------------------------------------------------------------|----------------------------------------|
| Noise bandwidth                                                      | 4 MHz                                  |
| Number of carriers                                                   | Analogue (AM-VSB)<br>$K = 57$ channels |
| Total modulation index                                               | 0,264                                  |
| $R/N$ of optical signal at the input of V-ONU                        | $-148 \text{ dB}(\text{Hz}^{-1})$      |
| Response of V-ONU                                                    | 0,89 A/W                               |
| Dark current of V-ONU                                                | 0,1 nA                                 |
| Equivalent input noise current density of pre-amplifier before V-ONU | $7 \text{ pA}/\sqrt{\text{Hz}}$        |

NOTE  $R/N$  of optical signal at the input of V-ONU is calculated when  $R/N$  of optical transmitter is  $-155 \text{ dB}(\text{Hz}^{-1})$ ,  $NF$  of optical amplifier is 6,5 (single stage, optical input 0 dB(mW)). Then the  $R/N$  due to optical transmission line is  $-161 \text{ dB}(\text{Hz}^{-1})$ .

If the optical modulation index of all the carriers is assumed to be the same, then the optical modulation index per carrier is given by,

$$m_k = \frac{0,264}{\sqrt{57}} \cong 0,035$$

If the optical input to the V-ONU is  $-9,6 \text{ dB}(\text{mW})$ , from Equation (3), the carrier-to-noise-ratio is calculated to be 43,0 dB.

## 6.5 Optical modulation index

The optical modulation index ( $OMI$ ) of analogue signals shall be measured according to the method described in 4.9 of IEC 60728-6. In this standard it is assumed that the power AGC function in the transmitter shall be off.

## 6.6 Carrier-to-crosstalk ratio (CCR)

### 6.6.1 General

This method of measurement is applicable when other services (i.e. digital communication signals like GPON, GEAPON or Ethernet-point-to-point) besides CATV broadcast transmission (i.e. AM-VSB and 64/256QAM broadcast signals) are transmitted in the optical network. Other services may produce crosstalk effects in optical fibres and in optical receiver devices with high linearity.

Crosstalk effects may arise when other service transmissions are applied by wavelength division multiplexing (WDM) on the same fibre and there is either insufficient optical wavelength filtering or relevant presence of non-linear fibre optical effects or both. Insufficient optical wavelength filtering may be due to low triplexer quality. Important non-linear fibre optical effects may be stimulated Raman scattering (SRS), self-phase modulation (SPM) and cross phase modulation (XPM). Among these causes SRS induced crosstalk is typically dominant, when analogue CATV broadcasting is transmitted on 1 550 nm wavelength and digital service signals use the 1 490 nm wavelength due to the fixed wavelength spacing.

### 6.6.2 Equipment

The following equipment is required:

- running system with implemented CATV broadcast service and other service(s);
- a selective voltmeter (or spectrum analyser) covering the frequency range of CATV broadcast service;

- sufficient length of fibre for connecting the transmitters, optical WDM filters, optical amplifiers, optical attenuators, optical polarization state change systems and receivers.

### 6.6.3 General measurements

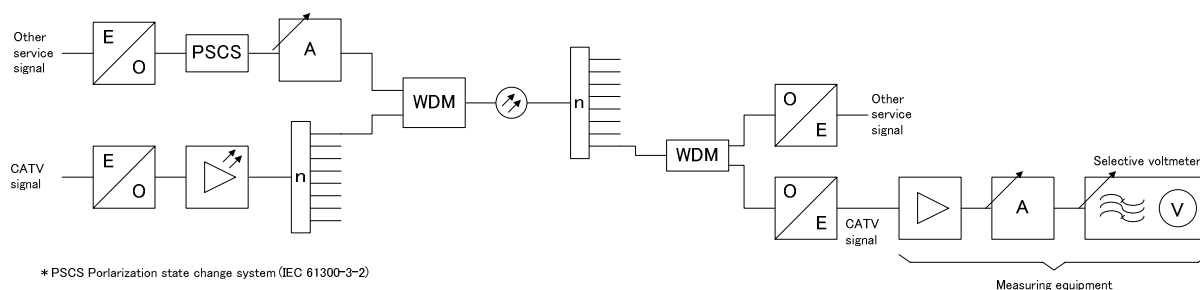
The following measurements are required:

- unless otherwise required, the reference levels used in the measurements shall be the normal operating levels;
- where the receiver to be measured includes automatic level control (ALC) pilot signals of the correct type, frequency and level shall be maintained throughout the tests.

### 6.6.4 Procedure

Proceed as follows.

- Set the supply voltage(s) and any control input signal(s) to the specified value(s).
- Connect the equipment as shown in Figure 11.



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**Figure 11 – Arrangement of test equipment for measuring other services crosstalk**

- Carry out measurements with the service signals in operation widely and closely spaced over each band of interest.
  - Carry out measurements over the full specified range, optical power at optical fibre input by adjusting optical attenuators A.
  - Carry out measurements over the full specified range of optical transmission distance by applying various fibre lengths. Figure C.6 shall be referred to.
  - Carry out measurements with various other services' communication signal patterns. For example measurements should be performed with and without digital idle signals (with and without payload), because signal pattern characteristics will influence crosstalk intensity.
  - Connect the variable RF attenuator A and selective voltmeter to the RF output port of optical receiver for CATV broadcast service. Tune the meter to each CATV carrier and note the attenuator A value  $a_1$  required to obtain a convenient meter reading  $R$  for the reference signal. The attenuator value  $a_1$  should be slightly greater than other services crosstalk CCR expected at the point of measurement.
  - Tune the meter to the other services' crosstalk product to be measured and tune the optical polarization state change system (PSCS) to the other services' crosstalk product to be measured at maximum level. Reduce the RF attenuator A by setting to the value  $a_2$  required to obtain the same meter reading  $R$ .
- NOTE It may be necessary to temporarily switch off one CATV carrier occupying the frequency band of local interest during measurements of other services' crosstalk single frequency level in order to obtain an accurate value  $a_2$ .
- The other services' crosstalk, in dB, is given by

$$CCR = a_1 - a_2$$

where

- $a_1$  is the RF attenuator A value when measuring the test signal used as reference, in dB;
- $a_2$  is the RF attenuator A value when measuring the crosstalk product, in dB.

### 6.6.5 Potential sources of error

Sources of error are the following:

- the inaccuracy of the selective voltmeter;
- the inaccuracy of the variable attenuators.

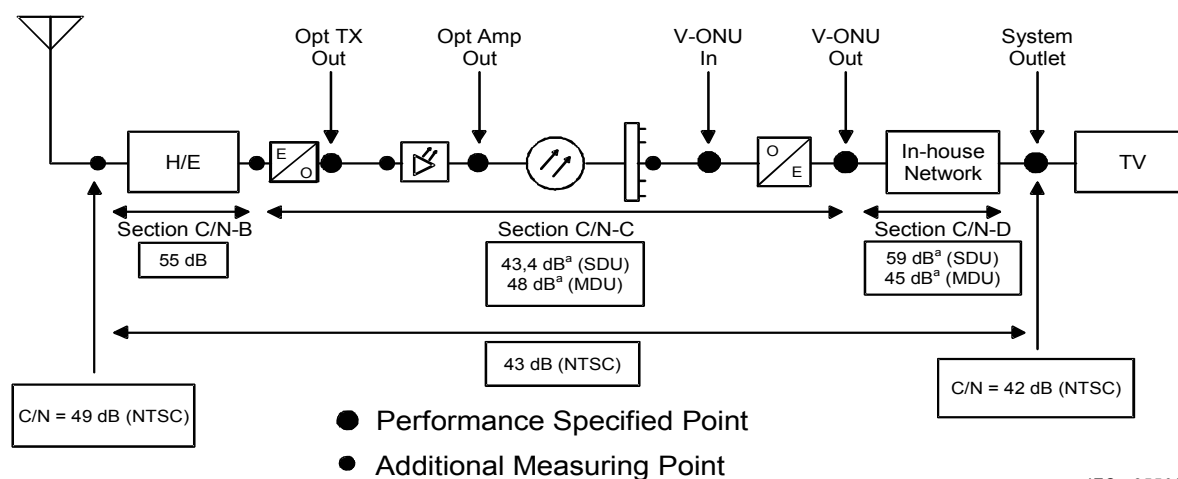
### 6.6.6 Presentation of the results

The other services' crosstalk (*CCR*) shall be expressed in dB.

## 7 Specification of optical system for broadcast signal transmission

### 7.1 Analogue and digital broadcast system over optical network

NTSC, PAL or SECAM systems are commonly used in analogue broadcast system with AM-VSB modulation systems. In digital broadcast systems over optical networks the 64 QAM system and the OFDM system are mainly used. The minimum *C/N* ratio at the headend input and system outlet for NTSC, PAL or SECAM systems are indicated in Table 5. These values shall be obtained on section *C/N* indicated in Figure 12, where the values for the NTSC system only are showed. For digitally modulated signals the Bit Error Ratio (*BER*) must be used as the specification parameter at system outlet:  $1 \times 10^{-4}$  is required for 64 QAM signals while  $2 \times 10^{-4}$  for OFDM signals. Figure 12 also depicts the measuring points (performance specified points) in order to guarantee the operating performance of the optical system. Additional measuring points are indicated for maintenance purposes. The measuring methods of Clauses 5 and 6 shall be used.



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<sup>a</sup> Appropriate optical level at V-ONU input shall be considered in the system design for MDU and SDU systems in the in-house network.

NOTE 1 All the values indicate the minimum requirement.

NOTE 2 Section C/N-D is not identical to the home network defined in IEC 60728-1.

NOTE 3 In the case of 51 dB *C/N* (IEC 60728-1-1) for Section C/N-D, the *C/N* for Section C/N-C should be 44 dB.

**Figure 12 – Performance allocation and measuring points**

For other system parameters Table 5 shall be referred to.

## 7.2 International TV systems

Table 5 and Table 6 can be applied to the FTTH systems covered by this standard for commercial use.

**Table 5 – Minimum C/N requirements in operation**

| Systems                    | Modulation   | Minimum C/N at headend input<br>dB | Minimum C/N at system outlet<br>dB |
|----------------------------|--------------|------------------------------------|------------------------------------|
| NTSC                       | AM-VSB<br>FM | 49<br>20                           | 42<br>13                           |
| I<br>B, G, D1<br>L<br>D, K | AM-VSB       | 50<br>50<br>48,5<br>50             | 43<br>43<br>44,5<br>43             |
| PAL<br>SECAM               | FM           | 21                                 | 14                                 |

**Table 6 – Minimum RF signal-to-noise ratio requirements in operation**

| System         | Modulation                            | Code rate                                                                  | Minimum RF signal-to-noise ratio at headend input<br>$S_{D,RF}/N$<br>dB                |                                                                               |                                                                                 |                                                                              | Minimum RF signal-to-noise ratio at system outlet<br>$S_{D,RF}/N$<br>dB             |                                                                              |                                                                                 |                                                                              |
|----------------|---------------------------------------|----------------------------------------------------------------------------|----------------------------------------------------------------------------------------|-------------------------------------------------------------------------------|---------------------------------------------------------------------------------|------------------------------------------------------------------------------|-------------------------------------------------------------------------------------|------------------------------------------------------------------------------|---------------------------------------------------------------------------------|------------------------------------------------------------------------------|
| DVB-S          | QPSK                                  | 1/2<br>2/3<br>3/4<br>5/6<br>7/8                                            | 8,6<br>10,5<br>11,5<br>12,6<br>13,3                                                    |                                                                               |                                                                                 |                                                                              | 6,6<br>8,5<br>9,6<br>10,6<br>11,3                                                   |                                                                              |                                                                                 |                                                                              |
| DVB-S2         | QPSK<br>8PSK<br>16APSK<br>32APSK<br>c | 1/4<br>1/3<br>2/5<br>1/2<br>3/5<br>2/3<br>3/4<br>4/5<br>5/6<br>8/9<br>9/10 | QPSK<br>3,7<br>4,8<br>5,7<br>7,0<br>8,2<br>9,1<br>10,0<br>10,7<br>11,2<br>12,2<br>12,4 | 8PSK<br>–<br>–<br>–<br>–<br>11,5<br>12,6<br>13,9<br>–<br>15,4<br>16,7<br>17,0 | 16APSK<br>–<br>–<br>–<br>–<br>–<br>15,0<br>16,2<br>17,0<br>17,6<br>18,9<br>19,1 | 32APSK<br>–<br>–<br>–<br>–<br>–<br>–<br>18,7<br>19,6<br>20,3<br>21,2<br>22,1 | QPSK<br>1,7<br>2,8<br>3,7<br>5,0<br>6,2<br>7,1<br>8,0<br>8,7<br>9,2<br>10,2<br>10,4 | 8PSK<br>–<br>–<br>–<br>–<br>9,5<br>10,6<br>11,9<br>–<br>13,4<br>14,7<br>15,0 | 16APSK<br>–<br>–<br>–<br>–<br>–<br>13,0<br>14,2<br>15,0<br>15,6<br>16,9<br>17,1 | 32APSK<br>–<br>–<br>–<br>–<br>–<br>–<br>16,7<br>17,6<br>18,3<br>19,4<br>20,1 |
| DVB-C          | 16 QAM<br>64 QAM<br>256 QAM           |                                                                            | 25,9<br>31,9<br>37,9                                                                   |                                                                               |                                                                                 |                                                                              | 19 <sup>a</sup><br>25 <sup>a</sup><br>31 <sup>a</sup>                               |                                                                              |                                                                                 |                                                                              |
| DVB-T<br>COFDM | QPSK<br>b                             | 1/2                                                                        | 2k mode and 8k mode                                                                    |                                                                               |                                                                                 |                                                                              | 2k mode                                                                             |                                                                              | 8k mode                                                                         |                                                                              |
|                |                                       | 2/3                                                                        | 6,1<br>8,2<br>9,3<br>10,5<br>11,3                                                      |                                                                               |                                                                                 |                                                                              | 4,9<br>7,2<br>8,5<br>9,9<br>10,9                                                    |                                                                              | 5,1<br>7,4<br>8,6<br>10,0<br>11,0                                               |                                                                              |
|                |                                       | 3/4                                                                        |                                                                                        |                                                                               |                                                                                 |                                                                              |                                                                                     |                                                                              |                                                                                 |                                                                              |
|                |                                       | 5/6<br>7/8                                                                 |                                                                                        |                                                                               |                                                                                 |                                                                              |                                                                                     |                                                                              |                                                                                 |                                                                              |
|                | 16 QAM<br>b                           | 1/2                                                                        | 2k mode and 8k mode                                                                    |                                                                               |                                                                                 |                                                                              | 2k mode                                                                             |                                                                              | 8k mode                                                                         |                                                                              |
|                |                                       | 2/3                                                                        | 12,2<br>14,2<br>15,6<br>17,1<br>17,7                                                   |                                                                               |                                                                                 |                                                                              | 11,0<br>13,2<br>14,7<br>16,4<br>17,3                                                |                                                                              | 11,2<br>13,4<br>14,9<br>16,6<br>17,3                                            |                                                                              |
|                |                                       | 3/4                                                                        |                                                                                        |                                                                               |                                                                                 |                                                                              |                                                                                     |                                                                              |                                                                                 |                                                                              |
|                |                                       | 5/6<br>7/8                                                                 |                                                                                        |                                                                               |                                                                                 |                                                                              |                                                                                     |                                                                              |                                                                                 |                                                                              |

| System                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | Modulation  | Code rate                       | Minimum RF signal-to-noise ratio<br>at headend input        |        |        | Minimum RF signal-to-noise ratio<br>at system outlet |                                                 |        |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------|---------------------------------|-------------------------------------------------------------|--------|--------|------------------------------------------------------|-------------------------------------------------|--------|
|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |             |                                 | $S_{D,RF}/N$<br>dB                                          |        |        | $S_{D,RF}/N$<br>dB                                   |                                                 |        |
|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | 64 QAM<br>b | 1/2<br>2/3<br>3/4<br>5/6<br>7/8 | 2k mode and 8k mode<br>17,4<br>20,0<br>21,6<br>23,3<br>24,5 |        |        | 2k mode<br>16,1<br>19,0<br>20,7<br>22,5<br>23,8      | 8k mode<br>16,3<br>19,2<br>20,9<br>22,6<br>23,9 |        |
| ISDB-T<br>(OFDM)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | 16QAM       | 1/2                             | Mode 1                                                      | Mode 2 | Mode 3 | Mode 1                                               | Mode 2                                          | Mode 3 |
|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |             | 2/3                             | d                                                           | d      | d      | 24                                                   | 24                                              | 24     |
|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |             | 3/4                             |                                                             |        |        | 24                                                   | 24                                              | 24     |
|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |             | 5/6                             |                                                             |        |        | 24                                                   | 24                                              | 24     |
|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |             | 7/8                             |                                                             |        |        | 24                                                   | 24                                              | 24     |
|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | 64QAM       | 1/2                             | Mode 1                                                      | Mode 2 | Mode 3 | Mode 1                                               | Mode 2                                          | Mode 3 |
|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |             | 2/3                             | d                                                           | d      | d      | 24                                                   | 24                                              | 24     |
|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |             | 3/4                             |                                                             |        |        | 24                                                   | 24                                              | 24     |
|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |             | 5/6                             |                                                             |        |        | 24                                                   | 24                                              | 24     |
|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |             | 7/8                             |                                                             |        |        | 24                                                   | 24                                              | 24     |
|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | QPSK        | 1/2                             | Mode 1                                                      | Mode 2 | Mode 3 | Mode 1                                               | Mode 2                                          | Mode 3 |
|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |             | 2/3                             | d                                                           | d      | d      | 24                                                   | 24                                              | 24     |
|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |             | 3/4                             |                                                             |        |        | 24                                                   | 24                                              | 24     |
|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |             | 5/6                             |                                                             |        |        | 24                                                   | 24                                              | 24     |
|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |             | 7/8                             |                                                             |        |        | 24                                                   | 24                                              | 24     |
|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | DQPSK       | 1/2                             | Mode 1                                                      | Mode 2 | Mode 3 | Mode 1                                               | Mode 2                                          | Mode 3 |
|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |             | 2/3                             | d                                                           | d      | d      | 24                                                   | 24                                              | 24     |
|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |             | 3/4                             |                                                             |        |        | 24                                                   | 24                                              | 24     |
|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |             | 5/6                             |                                                             |        |        | 24                                                   | 24                                              | 24     |
|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |             | 7/8                             |                                                             |        |        | 24                                                   | 24                                              | 24     |
| <p>a The above values take into account simultaneous distribution of analogue and digital signals. These values assume that intermodulation noise is not present or can be neglected and a <i>BER</i> of <math>10^{-4}</math> before Reed-Solomon decoder is achieved. For CATV networks intermodulation has to be considered in the time domain as clipping noise and a margin of 6 dB should be added even if the signal is regenerated in the headend.</p> <p>b These values take into account white noise and impulse noise.</p> <p>c These values are calculated according to ETSI EN 302 307, Tables 13 and H.1, and are intended for a <i>PER</i> of <math>10^{-7}</math> after LDPC and BCH decoders.</p> <p>d Every value is defined in <i>BER</i> <math>1 \times 10^{-4}</math> before Reed-Solomon decoder, not S/N.</p> |             |                                 |                                                             |        |        |                                                      |                                                 |        |
| Remark: The values for the system outlet were taken from IEC 60728-1-2, first edition The values for the headend input were calculated using the system outlet values and the system performance specified in Table 11 of IEC 60728-1-2.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |             |                                 |                                                             |        |        |                                                      |                                                 |        |

### 7.3 Relationship between *RIN* and *C/N*

The CATV broadcast service can be classified into three types:

- multi-channel service with a mixture of analogue and digital signals,
- digital only multi-channel service and
- re-transmission service for poor signal reception.

Table 7 shows the types of broadcast services with the typical number of carriers. For combination of analogue and digital carriers in actual system design, Annex A should be referred to.

**Table 7 – Types of broadcast services**

|                                                                       | Analogue (NTSC) system | Digital system              |
|-----------------------------------------------------------------------|------------------------|-----------------------------|
| a) Multi-channel service with mixture of analogue and digital signals | 40 carriers            | 30 carriers (64 QAM, OFDM)  |
| b) Digital only multi-channel service                                 | –                      | 110 carriers (64 QAM, OFDM) |
| c) Re-transmission service for poor signal reception                  | 10 carriers            | 10 carriers (OFDM)          |

Performance of transmission line can be defined by the value of relative intensity noise (*RIN*) for optical signal, and the *C/N* ratio at V-ONU output for electrical signal. The term V-ONU is used as the synonym of optical receiver (O/E) device in this document. Details on *RIN* measurement are described in Annex A. *RIN* values required for the three service types are shown in Table 8. The Intensity Modulation method is applied to the optical system in all service types.

**Table 8 – Type of service and minimum operational *RIN* values**

| Type of service                                                    | V-ONU minimum input level<br>dB(mW) | System <i>RIN</i> minimum value<br>dB(Hz <sup>-1</sup> ) | Corresponding <i>C/N</i> value<br>dB |
|--------------------------------------------------------------------|-------------------------------------|----------------------------------------------------------|--------------------------------------|
| Multi-channel service with mixture of analogue and digital signals | –8                                  | –149                                                     | 44                                   |
| Digital only multi-channel service                                 | –12                                 | –146                                                     | 28                                   |
| Re-transmission service for poor signal reception                  | –5                                  | –134                                                     | 44                                   |
|                                                                    | –8                                  | –134                                                     | 44                                   |
|                                                                    | –10                                 | –136                                                     | 44                                   |

Multiple optical reflections over transmission line may degrade *RIN* values. In order to minimize this degradation, the use of Grade 2 connectors of IEC 61755-1 or APC optical connectors is recommended.

#### 7.4 Optical wavelength

In accordance with ITU-T Recommendation G.983.3, a wavelength within the range of 1 555 nm ± 5 nm is strongly recommended for the combined system of broadcast and telecommunication systems over single fibre. For DWDM systems wavelengths according to the grid defined in ITU-T Recommendation G.692 shall be used.

In CWDM systems ITU-T Recommendation G.694.2 shall be taken into account.

NOTE Considering the availability of EDFA and appropriate allocation of wavelengths, the wavelength range of 1 530 nm to 1 625 nm (edge of C-band and L-band of ITU-T Recommendation G.694.2) can be used for actual and economical video transmission.

#### 7.5 Frequency of source signal

The frequency range of source signals considered here is 47 MHz to 862 MHz. However, regional frequency plans can be used for the operating frequency range of the optical system.

#### 7.6 Optical system specification for broadcast signal transmission

The specification of the most important parameters of the optical system is shown in Table 9.

**Table 9 – Optical system specification**

| Parameter                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |                                  | Analogue TV<br>NTSC                                        | Analogue TV<br>PAL, SECAM | Digital TV<br>64QAM <sup>d</sup>       | Digital TV<br>OFDM |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------|------------------------------------------------------------|---------------------------|----------------------------------------|--------------------|
| Optical wavelength                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |                                  | 1 530 nm to 1 625 nm                                       |                           |                                        |                    |
| Frequency of source signal                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |                                  | 90 MHz to 770 MHz (NTSC)<br>47 MHz to 862 MHz (PAL, SECAM) |                           |                                        |                    |
| Fluctuation of carrier-wave level                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |                                  | Within 4 dB for<br>a duration of<br>1 min <sup>b</sup>     | –                         | Within 3 dB<br>for a duration of 1 min |                    |
| Noise bandwidth of electronic signal                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |                                  | 4 MHz                                                      | 5 MHz <sup>c</sup>        | 5,3 MHz                                | 5,6 MHz            |
| Minimum section <i>C/N</i><br>(Between head-end input and<br>system outlet)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |                                  | 44 dB                                                      | 46,5 dB                   | 28 dB                                  | 27 dB              |
| Minimum <i>C/N</i>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | At head-end<br>input             | 51 dB                                                      | 53 dB                     | –                                      | –                  |
|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | At system<br>outlet <sup>a</sup> | 43 dB                                                      | 44,5 dB                   | 26 dB                                  | 24 dB              |
| Minimum <i>BER</i>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | At head-end<br>input             | –                                                          | –                         | $1 \times 10^{-4}$                     | $1 \times 10^{-4}$ |
|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | At system<br>outlet <sup>a</sup> | –                                                          | –                         | $1 \times 10^{-4}$                     | $2 \times 10^{-4}$ |
| <sup>a</sup> Minimum <i>C/N</i> must be specified at system outlet. However, it may allow to specify the <i>C/N</i> at the input of V-ONU which is the interface point to customer premises, as an alternative way. In such a case additional 2 dB are required to the minimum <i>C/N</i> in Table 9. Minimum <i>BER</i> shall remain unchanged in Table 9.<br><sup>b</sup> Such transients may be observed in the electrical section of Headend which has AGC functions to absorb unexpected fluctuations including fading, and such a fluctuation is a rare event, not a permanent one.<br><sup>c</sup> Noise bandwidth of each system shall be referred to Table 4.<br><sup>d</sup> ITU-T Recommendation J.83, Annex C. |                                  |                                                            |                           |                                        |                    |

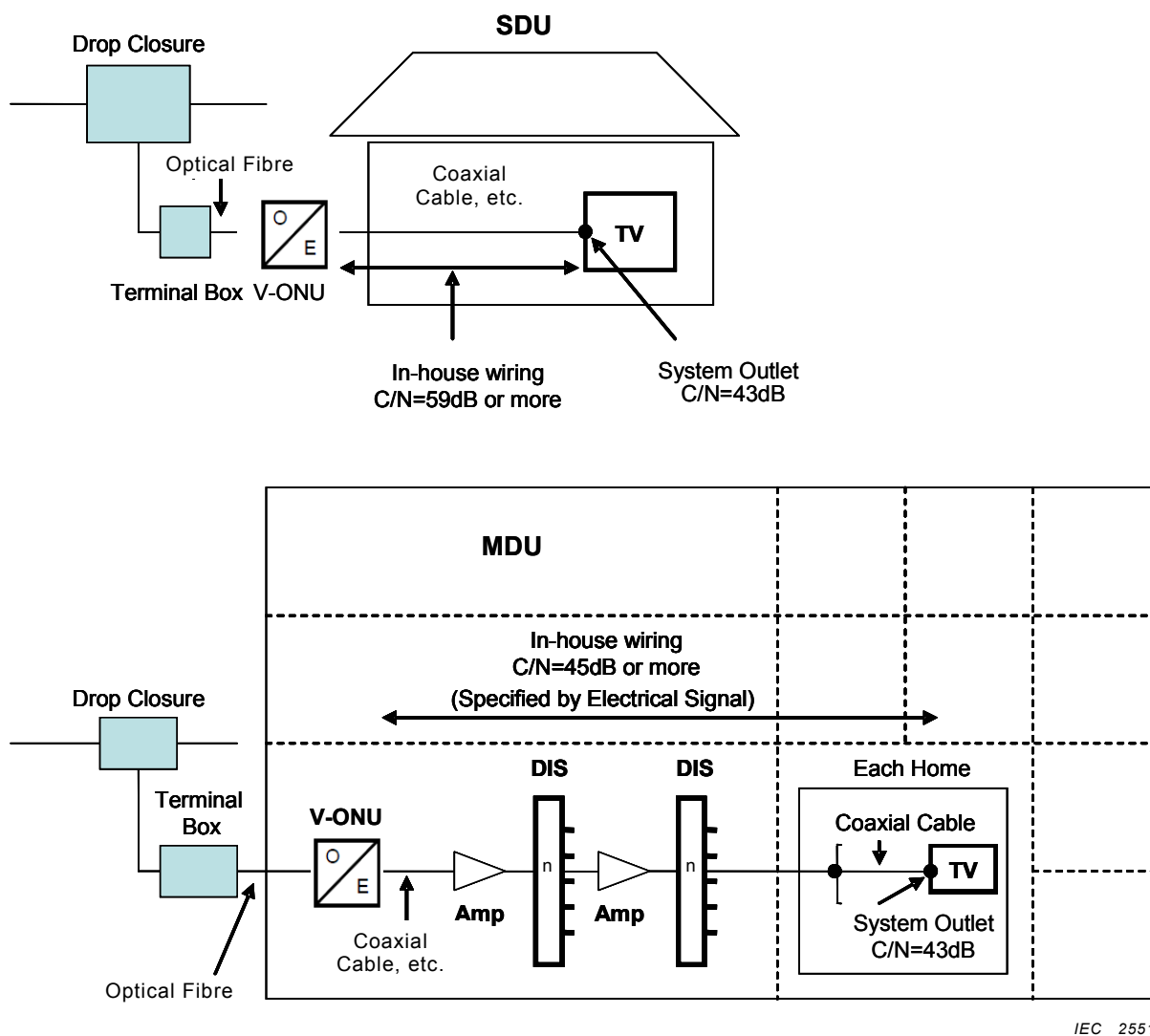
### 7.7 *C/N* ratio specification for in-house and in-building wirings

The *C/N* ratio can be specified outside of the system outlet if the performance of the in-house/in-building wiring section is maintained properly. Based on current installation methods, *C/N* ratio allocation for the in-house/in-building wiring section is specified in Table 10. *C/N* ratio allocation is different between Single Dwelling Unit (SDU) and Multiple Dwelling Unit (MDU) as shown in Figure 13 and Figure 14. The coaxial distribution network after V-ONU shall be designed in such a manner that the *C/N* ratio of in-house wiring should be not lower than 59 dB in SDU, while it should be not lower than 45 dB in MDU, taking into account the *C/N* degradation due to the distribution network only, therefore assuming an unimpaired input signal to the in-home distribution network.

**Table 10 – Section of C/N ratio specification for in-house/in-building wiring**

| Category of house            |                                                                                         | Minimum C/N ratio for in-house/ in-building wiring dB |
|------------------------------|-----------------------------------------------------------------------------------------|-------------------------------------------------------|
| Single dwelling unit (SDU)   | –                                                                                       | 59                                                    |
| Multiple dwelling unit (MDU) | O/E conversion at MDU entrance, coaxial cable distribution to TV set (see Figure 13)    | 45                                                    |
|                              | No O/E conversion at MDU entrance, optical cable distribution to TV set (see Figure 14) | 45                                                    |

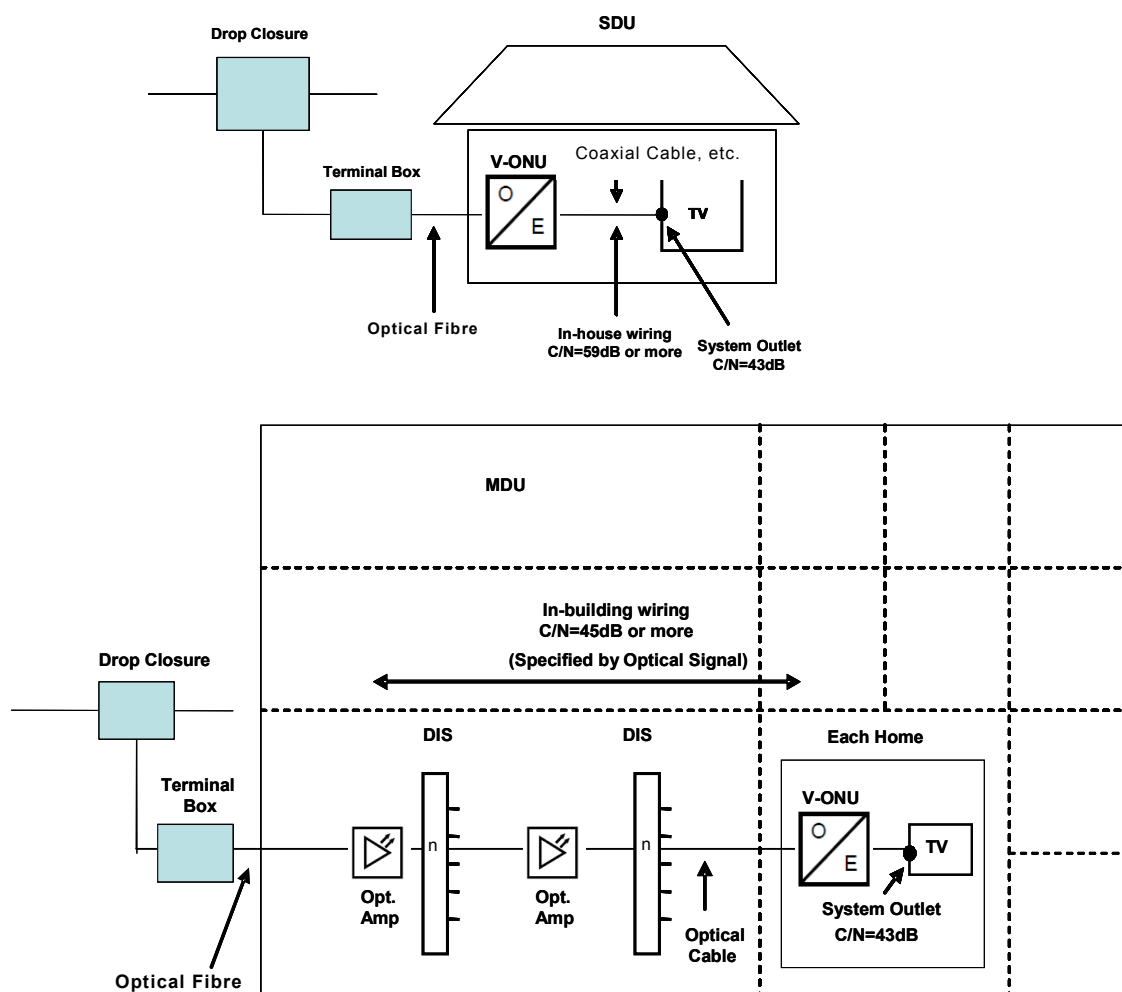
NOTE The 59 dB is not a requirement for the IEC 60728-1 home-network interface, but a practical specification for the RF amplifier that is built-in the optical receiver of the analogue RF wavelength of the PON.



IEC 2551/09

**Figure 13 – Section of C/N ratio specification (45 dB) for in-house wiring (specified for electrical signals)**





**Figure 14 – Section of C/N ratio specification for in-house wiring  
(specified for optical signals)**

IEC 2552/09

## 7.8 Crosstalk due to optical fibre non-linearity

In a hybrid WDM transmission system in which broadcast signal and telecommunication signal are both incorporated, crosstalk between the two signals due to optical fibre non-linearity shall be taken into consideration. As described in Clause D.2, important parameters of optical crosstalk are stimulated Raman scattering (SRS), self-phase modulation (SPM), and Cross phase modulation (XPM). Among these parameters only SRS-induced crosstalk is dominant and it is difficult to reduce interference based on the Raman effect (change of optical wavelength) and the fixed wavelength spacing between the two signals (1 550 nm and 1 490 nm). XPM is not dominant in the case of this kind of wavelength spacing. SPM is caused by non-linear refractive index of fibre materials and reduced significantly when the input power is decreased.

Considering the above features of crosstalk, the following points shall be maintained as a minimum guideline for reducing SRS-induced crosstalk caused by optical fibre non-linearity.

- Optical level of telecommunication signal at trunk line fibre input: Less than 0 dB(mW).
- Optical level of broadcast signal at trunk line fibre input: Less than 18 dB(mW).
- Optical modulation index of broadcast signal: More than 3 %/Carrier, total optical modulation index shall be in accordance with the description in IEC/TR 60728-6-1.
- Randomization of telecommunication signal pattern: Recommended.

NOTE 1 Annex D contains additional descriptions of SRS.

NOTE 2 Table A.3 should be referred to for the description of the optical modulation index and the number of combinations of analogue and digital carriers.

## 7.9 Single frequency interference level due to fibre non-linearity

The single frequency interference level caused by fibre non-linearity shall meet the following values shown in Table 11. All the parameters of optical broadcast transmission systems shall be set appropriately to satisfy the interference level. The measuring points shall be point (6) shown in Figure 5.

**Table 11 – Interference level due to fibre non-linearity**

| Broadcast system | D/U ratio<br>dB |
|------------------|-----------------|
| AM-VSB (NTSC)    | More than 52    |
| 64QAM (Japan)    | More than 39    |
| 64QAM (DVB)      | More than 33    |
| OFDM (ISDB-T)    | More than 45    |

## 7.10 Environmental conditions

Equipment used for FTTH systems shall meet the environmental condition requirements of Table 12.

**Table 12 – Environmental conditions**

|                                                                                                                        | Optical Tx<br>(In-house) | EDFA<br>(In-house) | V-ONU<br>(In-house) | V-ONU<br>(Outdoor)         |
|------------------------------------------------------------------------------------------------------------------------|--------------------------|--------------------|---------------------|----------------------------|
| Temperature °C                                                                                                         | 0 to +40                 | 0 to +40           | 0 to +40            | –20 to +40<br>(Notes 1, 2) |
| Humidity %                                                                                                             | 20 to 90                 | 20 to 90           | 20 to 90            | 20 to 100                  |
| NOTE 1 Except the rising temperature due to solar radiation.                                                           |                          |                    |                     |                            |
| NOTE 2 To be applied in normal climate conditions in extra-tropical zones, not in extremely cold and hot temperatures. |                          |                    |                     |                            |
| NOTE 3 Refer to IEC 60068-1, IEC 60050-191 and ETSI 3000019-1-4 when used in other climate conditions.                 |                          |                    |                     |                            |

## Annex A (informative)

### Actual service systems and design considerations

#### A.1 Multi-channel service system

##### A.1.1 General

This annex describes actual service systems and design considerations based on the specifications stated in this standard.

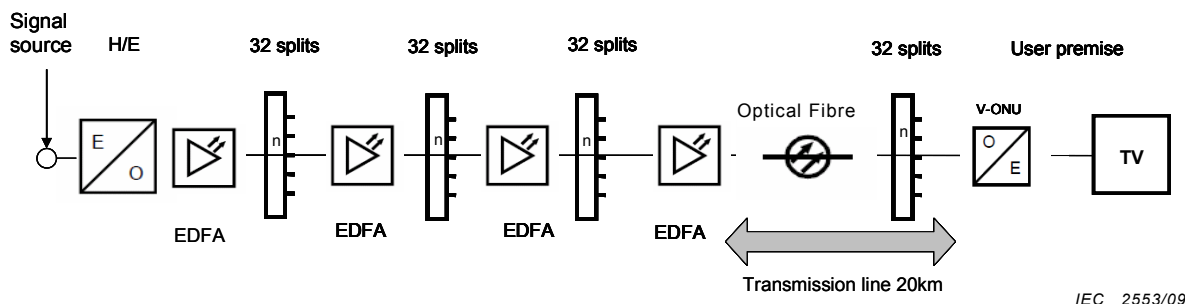
CATV operator currently provides multi-channel services for mixed analogue and digital broadcasting. Analogue and/or digital multi-channel services are classified as multi-channel services in this standard. Since the actual number of carriers in CATV network varies, this annex describes the following two reference models for the multi-channel services over an optical network.

(1) Analogue 57 carriers + Digital 40 carriers

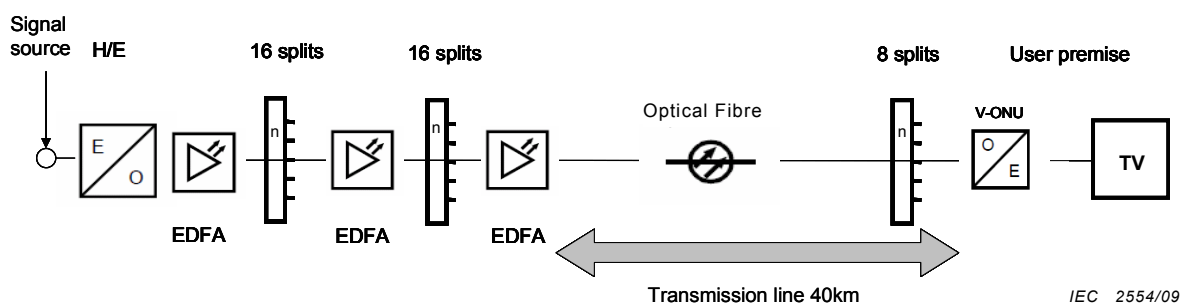
(2) Analogue 11 carriers + Digital 80 carriers

Model (1) is selected as a reference model that has the maximum number of both analogue and digital transmission carriers at present. Model (2) is the reference model that can carry up to 80 digital carriers, with a small number of analogue carriers. In both cases, the maximum number of carriers is assumed. “Analogue” here means NTSC signals, and “digital”, either 64 QAM or OFDM signals, regardless of the presence of digital carriers. Refer to Clause H.1 of IEC 60728-1.

The system size of CATV multi-channel services is mostly from 1 500 to 340 000 terminals, the transmission line length is up to 40 km, and 4 stages of EDFA in general. Term EDFA is the synonym of optical amplifier in this standard. In the optical network system, the relationship between transmission distance and the number of branches is in inverse proportion. If the number of branches is reduced, optical line is extendable up to 40 km, and one million or more subscriber terminals become available by stacking splitters. Figure A.1 and Figure A.2 show the examples of the multi-channel service system of one million terminals, and of 2 000 terminals, respectively.



**Figure A.1 – Example of a multi-channel service system of one million terminals**



**Figure A.2 – Example of a multi-channel service system of 2 000 terminals**

### A.1.2 Operating conditions

Table A.1 shows the two models of operating conditions for the multi-channel service system with its optical modulation index for each analogue or digital carrier. The optical system uses the external intensity modulation method.

**Table A.1 – Operating conditions of a multi-channel service system**

| System                                 | Number of carriers                            | Modulation method             | Optical modulation index ( $m_k$ )/Carrier |
|----------------------------------------|-----------------------------------------------|-------------------------------|--------------------------------------------|
| Multi-channel service system model (1) | Analogue: 57 carriers<br>Digital: 40 carriers | External intensity modulation | Analogue: 3,5 %<br>Digital: 1,1 %          |
| Multi-channel service system model (2) | Analogue: 11 carriers<br>Digital: 80 carriers |                               | Analogue: 7 %<br>Digital: 2,2 %            |

### A.1.3 Operating environment

The optical transmitter and optical amplifier are assumed to be installed in an office building as well as head end equipment, in the following environmental conditions. V-ONU is assumed to be installed indoors or outdoors like under eaves. Unless otherwise specified, the following range should be applied.

- a) Optical transmitter
 

|                     |                |                                       |
|---------------------|----------------|---------------------------------------|
| Ambient temperature | 0 °C to +40 °C |                                       |
| Humidity            | 20 % to 90 %   | without dew condensation <sup>a</sup> |
- b) Optical amplifier
 

|                     |                |                                       |
|---------------------|----------------|---------------------------------------|
| Ambient temperature | 0 °C to +40 °C |                                       |
| Humidity            | 20 % to 90 %   | without dew condensation <sup>a</sup> |
- c) V-ONU
 

|                     |                  |                                       |
|---------------------|------------------|---------------------------------------|
| Ambient temperature | –20 °C to +40 °C | (outdoor installation)                |
| Ambient temperature | 0 °C to +40 °C   | (indoor installation)                 |
| Humidity            | 20 % to 100 %    | without dew condensation <sup>a</sup> |

<sup>a</sup> Except the rising temperature due to solar radiation.

For optical safety, refer to IEC 60825-1 and IEC 60825-2 in addition to the relevant clauses of this standard.

## A.2 Re-transmission service system

### A.2.1 General

The re-transmission service system is a small-sized receiving facility with poor reception to measure broadcast TV programs, and does not support transmission of terrestrial digital broadcast signals. Nine carriers in each analogue and digital signal are assumed for the re-transmission service. General, re-transmission service systems have about 2 km trunk line and 70 terminals. In this case, one EDFA at its maximum is enough for this network size. Most of the re-transmission service system does not require an EDFA. It is enough to describe two types as shown in Figure A.3 and Figure A.4.

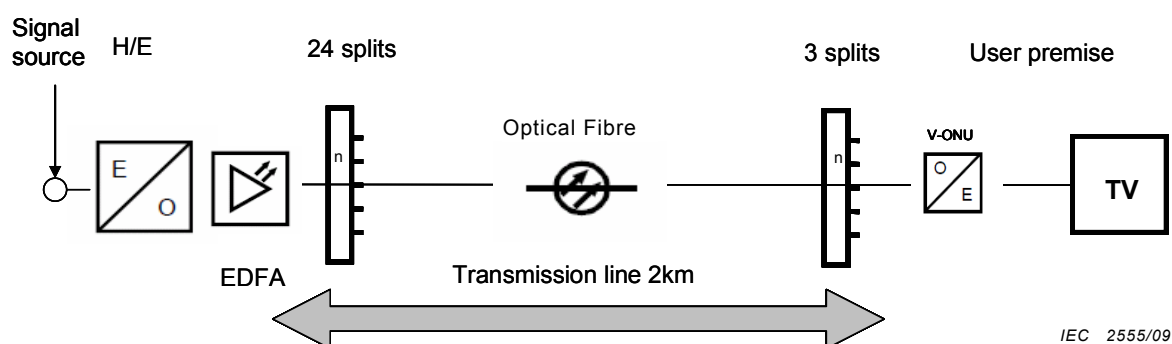


Figure A.3 – Example of re-transmission service system of 72 terminals

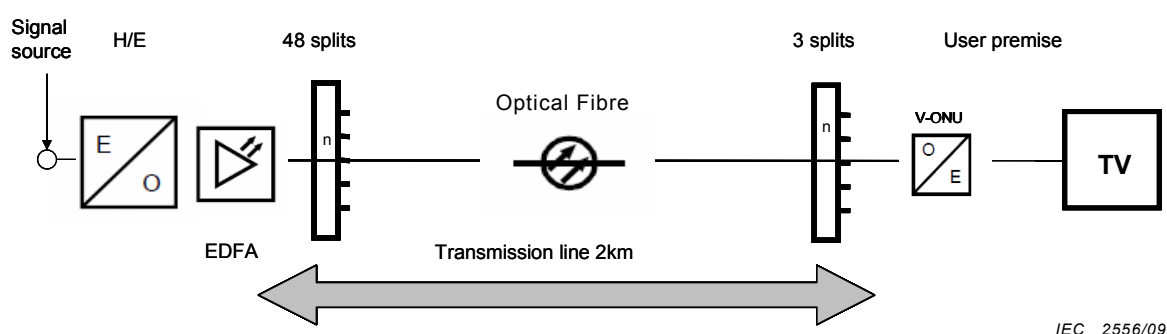


Figure A.4 – Example of re-transmission service system of 144 terminals

### A.2.2 Operating conditions

Table A.2 shows the model of operating conditions for the re-transmission service system with its optical modulation index. The optical system uses the direct intensity modulation method.

Table A.2 – Operating conditions of re-transmission service system

| System                         | Number of carriers                          | Modulation method           | Optical modulation index ( $m_k$ ) |
|--------------------------------|---------------------------------------------|-----------------------------|------------------------------------|
| Re-transmission service system | Analogue: 9 carriers<br>Digital: 9 carriers | Direct intensity modulation | Analogue: 9 %<br>Digital: 2,9 %    |

### A.2.3 Operating environment

The optical transmitter and optical amplifier are assumed to be installed in an office building with head-end equipment, in the following environmental conditions. V-ONU is supposed to be

installed indoors or outdoors like under eaves. Unless otherwise specified, the following range shall be applied.

- a) Optical transmitter
  - Ambient temperature –20 °C to +40 °C
  - Humidity 20 % to 90 % <sup>a</sup> without dew condensation
- b) Optical amplifier
  - Ambient temperature –20 °C to +40 °C
  - Humidity 20 % to 90 % <sup>a</sup> without dew condensation
- c) V-ONU
  - Ambient temperature –20 °C to +40 °C (outdoor installation)
  - Ambient temperature 0 °C to +40 °C (indoor installation)
  - Humidity 20 % to 100 % <sup>a</sup> without dew condensation

<sup>a</sup> Except the rising temperature due to solar radiation.

### A.3 C/N ratio calculation of optical network

In the case of optical intensity modulation, the C/N ratio of optical network can be calculated as follows.

- a) RIN degradation by optical amplifier

RIN degradation by multi-stage connection of the optical amplifiers is given by the following formula:

$$RIN_{out} = 10 \lg \left( \sum_k \frac{2 \cdot E \cdot 10^{\frac{NF_k}{10}}}{10^{\frac{P_k}{10}}} + 10^{\frac{RIN_{in}}{10}} \right) \text{ [dB(Hz}^{-1}\text{)]} \quad (\text{A.1})$$

where

- $RIN_{in}$  is the RIN at input of the first optical amplifier [dB(Hz<sup>-1</sup>)],
- $RIN_{out}$  is the RIN at input of the  $k$ -th optical amplifier [dB(Hz<sup>-1</sup>)],
- $E$  is the photon energy (in case of  $\lambda = 1\,555$  nm,  $E = 1,278 \times 10^{-16}$  [mJ]),
- $NF_k$  is the noise figure of the  $k$ -th optical amplifier [dB],
- $P_k$  is the optical input power of the  $k$ -th optical amplifier [dB(mW)].

- b) C/N ratio in the case of intensity modulation system

$$C/N = 10 \lg \left( \frac{1}{B_N} \cdot \frac{\frac{1}{2} \cdot (m_k \cdot R \cdot P_r)^2}{RIN(R \cdot P_r)^2 + 2 \cdot e \cdot (I_{d0} + R \cdot P_r) + I_{eq}^2} \right) \text{ [dB]} \quad (\text{A.2})$$

where

- $B_N$  is the noise bandwidth (AM-VSB:  $4,0 \times 10^6$  [Hz], 64 QAM:  $5,3 \times 10^6$  [Hz]),
- $m_k$  is the optical modulation index of the  $k$ -th carrier,
- $R$  is the optical-electrical conversion efficiency of an optical receiver device [A/W],

- $P_r$  is the input optical power [W],  
 $RIN$  is the relative intensity noise of optical input signal ( $RIN$ ) [1/Hz],  
 $e$  is the charge of electron ( $1,602 \times 10^{-19}$  [C]),  
 $I_{d0}$  is the dark current of a optical receiver device [A],  
 $I_{eq}$  is the equivalent input noise current density of an optical receiver [A/√Hz].

The total optical modulation index “ $M$ ” has the relationship with optical modulation index “ $m_k$ ” at the  $k$ -th carrier, and the number of transmission carriers “ $K$ ”, as shown in the following formula:

$$M = \sqrt{\sum_{k=1}^K m_k^2} \quad (\text{A.3})$$

Since the total optical modulation index affects the distortion characteristic of the total system, it is generally desirable to be 30 % or less in case of an external intensity modulation system.

#### A.4 System reference model

Table A.3 summarizes the basic system parameters verified for multi-channel and re-transmission service systems, the reference models are shown in Figure A.5 to Figure A.11.

**Table A.3 – Basic system parameters for multi-channel and re-transmission service systems**

| Reference model | Type                           | Number of carriers                            | Modulation index ( $m_k$ )        | EDFA stages | Remark             |
|-----------------|--------------------------------|-----------------------------------------------|-----------------------------------|-------------|--------------------|
| No. 1, 2        | Multi-channel service system   | Analogue: 57 carriers<br>Digital: 40 carriers | Analogue: 3,5 %<br>Digital: 1,1 % | 4 steps     | Figures A.5, A.6   |
|                 |                                |                                               |                                   | 3 steps     |                    |
| No. 3           |                                | Analogue: 40 carriers<br>Digital: 30 carriers | Analogue: 4,3 %<br>Digital: 1,4 % | 4 steps     | Figure A.7         |
| No. 4, 5        |                                | Analogue: 11 carriers<br>Digital: 80 carriers | Analogue: 7 %<br>Digital: 2,2 %   | 4 steps     | Figures A.8, A.9   |
|                 |                                |                                               |                                   | 4 steps     |                    |
| No. 6, 7        | Re-transmission service system | Analogue: 9 carriers<br>Digital: 9 carriers   | Analogue: 9 %<br>Digital: 2,9 %   | None        | Figures A.10, A.11 |
|                 |                                |                                               |                                   | 1 step      |                    |

The parameters used for calculation of system performance are as follows.

#### Multi-channel service system, reference models No. 1 to No. 5, Figures A.5 to A.9

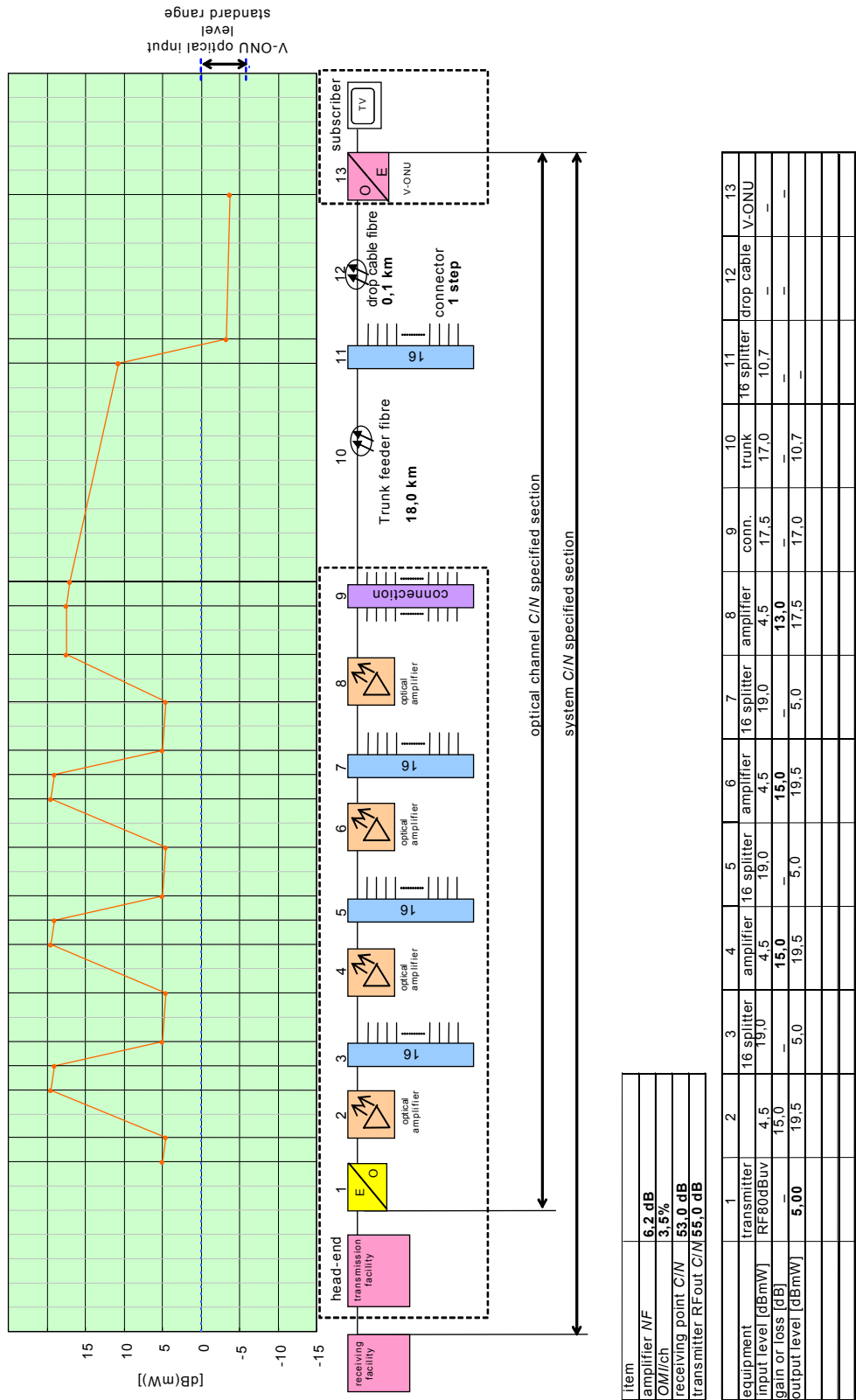
|                                                           |              |
|-----------------------------------------------------------|--------------|
| Connection loss at connector                              | 0,5 dB/point |
| Fibre loss including splicing loss                        | 0,35 dB/km   |
| V-ONU equivalent input noise current density ( $I_{eq}$ ) | 10 pA/√Hz    |
| V-ONU receiving device dark current ( $I_{d0}$ )          | 1,3 nA       |
| V-ONU optical-electrical conversion efficiency ( $R$ )    | 0,84 A/W     |

**Re-transmission service system, reference models No. 6 and No. 7,  
Figures A.10 and A.11**

|                                                           |                            |
|-----------------------------------------------------------|----------------------------|
| Connection loss at connector                              | 0,5 dB/point               |
| Fibre loss including splicing loss                        | 0,35 dB/km                 |
| V-ONU equivalent input noise current density ( $I_{eq}$ ) | 8,3 pA/ $\sqrt{\text{Hz}}$ |
| V-ONU receiving device dark current ( $I_{d0}$ )          | 1,0 nA                     |
| V-ONU optical-electrical conversion efficiency ( $R$ )    | 0,9 A/W                    |



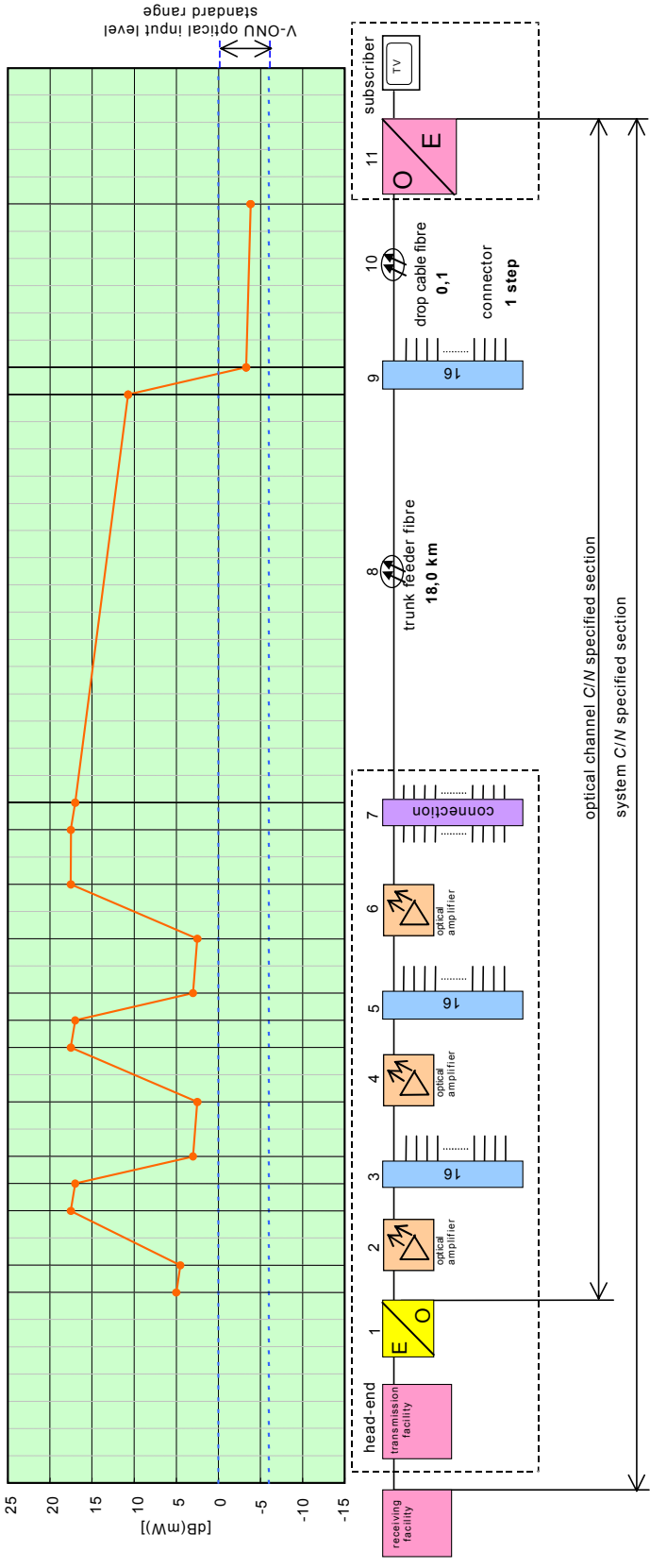
Model No.1 – Multi-channel service system, 4 steps of EDFA(s), OMI: analogue 3,5 % (57 carriers), digital 1,1 % (40 carriers)



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Figure A.5 – Model No.1 of a system performance calculation

Model No.2 – Multi-channel service system: 3 steps of EDFA(s); OMI: analogue 3,5 % (57 carriers), digital 1,1 % (40 carriers)



| item                  | value   |
|-----------------------|---------|
| amplifier NF          | 6,2 dB  |
| OMI /ch               | 3,5%    |
| receiving point C/N   | 53,0 dB |
| transmitter RFout C/N | 55,0 dB |

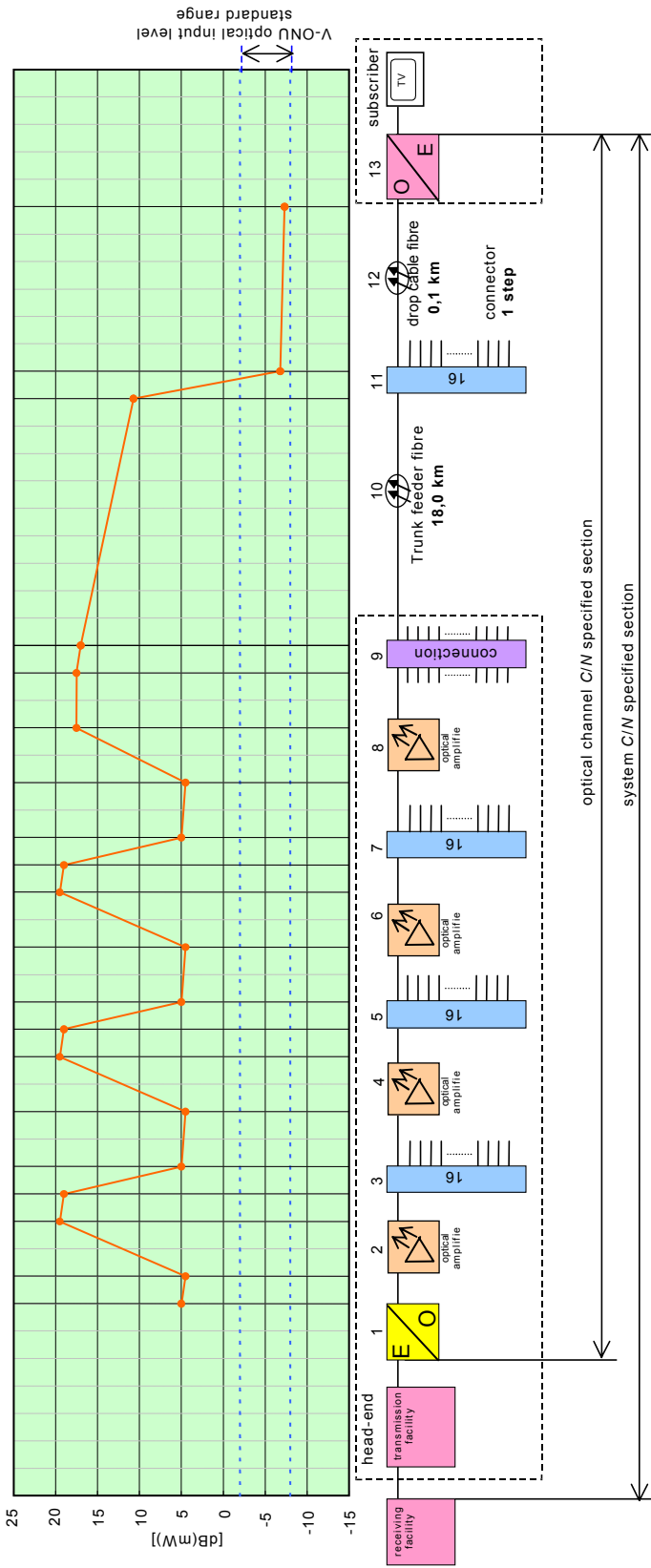
|                      |             |           |             |           |             |           |       |       |             |            |          |
|----------------------|-------------|-----------|-------------|-----------|-------------|-----------|-------|-------|-------------|------------|----------|
| equipment            | 1           | 2         | 3           | 4         | 5           | 6         | 7     | 8     | 9           | 10         | 11       |
| input level [dBmW]   | transmitter | amplifier | 16 splitter | amplifier | 16 splitter | amplifier | conn. | trunk | 16 splitter | drop cable | V-ONU    |
| gain or loss [dB]    | RF80dBuv    | 4,5       | 17,0        | 2,5       | 17,0        | 2,5       | 17,5  | 17,0  | 10,7        | -          | -        |
| output level [dBmW]  | -           | 13,0      | -           | 15,0      | -           | 15,0      | -     | -     | -           | -          | -        |
| output level [dBmW]  | 5,00        | 17,5      | 3,0         | 17,5      | 3,0         | 17,5      | 17,0  | 10,7  | -           | -          | RF80dBuv |
| output RIN[dB(Hz-1)] | -           | -         | -           | -         | -           | -         | -     | -     | -           | -          | -        |
| channel C/N [dB]     | -           | -         | -           | -         | -           | -         | -     | -     | -           | -          | 46,4     |
| system C/N [dB]      | -           | -         | -           | -         | -           | -         | -     | -     | -           | -          | 45,1     |

NOTE 1 optical channel C/N at the V-ONU RF output, which is calculated from R/N of the optical transmitter and the amplifier  
NOTE 2 C/N at V-ONU RF output, which is calculated from the receiving point C/N, transmitter C/N and optical channel C/N  
NOTE 3 connection loss between optical amplifier and connection panel is included in the loss of the connection panel

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Figure A.6 – Model No.2 of a system performance calculation

Model No.3 – Multi-channel service system: 4 steps of EDFA(s); OMI: analogue 4,3 % (40 carriers), digital 1,4 % (30 carriers)



| item                  | value   |
|-----------------------|---------|
| amplifier NF          | 6,2 dB  |
| OMI/ch                | 4,31%   |
| receiving point C/N   | 53,0 dB |
| transmitter RFout C/N | 55,0 dB |

|                                    |          |           |             |           |             |           |             |           |         |         |             |            |          |
|------------------------------------|----------|-----------|-------------|-----------|-------------|-----------|-------------|-----------|---------|---------|-------------|------------|----------|
| equipment                          | 1        | 2         | 3           | 4         | 5           | 6         | 7           | 8         | 9       | 10      | 11          | 12         | 13       |
| transmitter                        | RF80dBuV | amplifier | 16 splitter | amplifier | 16 splitter | amplifier | 16 splitter | amplifier | conn.   | trunk   | 16 splitter | drop cable | V-ONU    |
| input level [dBmW]                 |          | 4,50      | 19,00       | 4,50      | 19,00       | 4,50      | 19,00       | 4,50      | 17,50   | 17,00   | 10,70       | -6,80      | -7,34    |
| gain or loss [dB]                  |          |           | -14,00      | 15,00     | -14,00      | 15,00     | -14,00      | 13,00     | -0,50   | -6,30   | -17,50      | -0,54      | -        |
| output level [dBmW]                |          | 5,00      | 19,50       | 19,50     | 5,00        | 19,50     | 5,00        | 17,50     | 17,00   | 10,70   | -6,80       | -7,34      | RF80dBuV |
| output R/N [dB(Hz <sup>-1</sup> )] |          | -158,10   | -152,73     | -150,40   | -148,90     | -147,78   | -147,78     | -147,78   | -147,78 | -147,78 | -147,78     | -147,78    | -147,78  |
| channel C/N [dB]                   |          | -         | -           | -         | -           | -         | -           | -         | -       | -       | -           | -          | -        |
| system C/N [dB]                    |          | -         | -           | -         | -           | -         | -           | -         | -       | -       | -           | -          | 43,75    |

NOTE 1 optical channel C/N at the V-ONU RF output, which is calculated from R/N of the optical transmitter and the amplifier  
NOTE 2 C/N at V-ONU RF output, which is calculated from the receiving point C/N, transmitter C/N and optical channel C/N  
NOTE 3 connection loss between optical amplifier and connection panel is included in the loss of the connection panel

Figure A.7 – Model No.3 a of system performance calculation

Figure 1 illustrates a system C/N budget. The diagram shows a signal path from a transmitting facility through 13 steps to a subscriber. The path includes a head-end, a connection, and a subscriber section. The signal level is plotted in dB(mW) on the y-axis (from -15 to 25) against the step number on the x-axis (from 1 to 13). The signal level starts at approximately 15 dB(mW) at step 1, drops to 10 dB(mW) at step 2, rises to 12 dB(mW) at step 3, and then fluctuates between 10 and 15 dB(mW) until step 11. At step 12, it drops to -10 dB(mW) and remains there at step 13. A dashed blue line indicates the standard range from -10 to -5 dB(mW). A horizontal arrow at the bottom indicates the V-ONU optical input level.

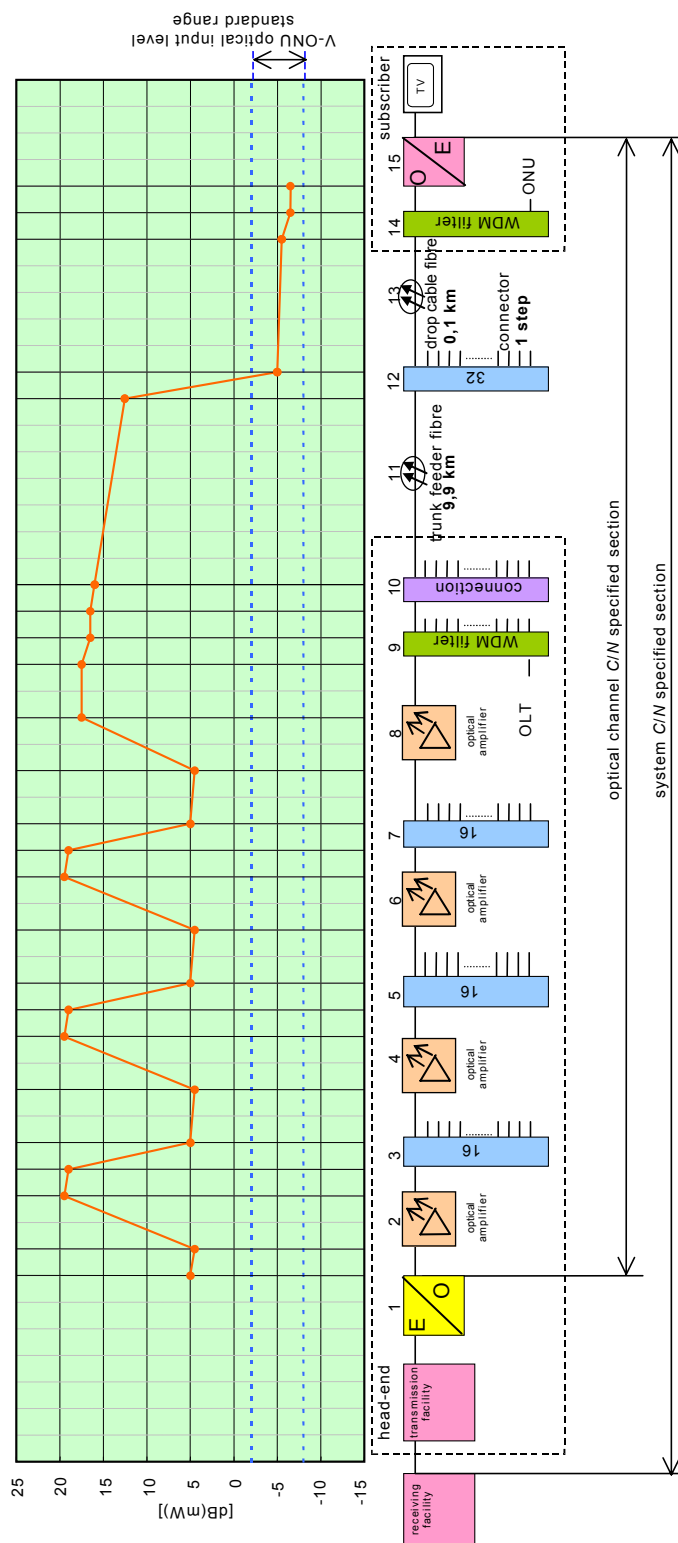
| item                  | value   |
|-----------------------|---------|
| amplifier NF          | 6,2 dB  |
| OMI/ch                | 7,0 % % |
| receiving point C/N   | 53,0 dB |
| Transmitter RFout C/N | 55,0 dB |

|                                    | 1              | 2            | 3           | 4            | 5           | 6            | 7           | 8            | 9     | 10    | 11          | 12         | 13       |
|------------------------------------|----------------|--------------|-------------|--------------|-------------|--------------|-------------|--------------|-------|-------|-------------|------------|----------|
| transmitter                        |                | amplifier    | 16 splitter | amplifier    | 16 splitter | amplifier    | 16 splitter | amplifier    | conn. | trunk | 32 splitter | drop cable |          |
| RF80dBuV                           |                | 4.50         | 17.00       | 2.50         | 17.00       | 2.50         | 17.00       | 2.50         | 17.50 | 17.00 | 10.70       | -6.80      | V-ONU    |
| input level [dBmW]                 |                |              |             |              |             |              |             |              |       |       |             |            | -7.34    |
| gain or loss [dB]                  |                | <b>13.00</b> | -14.00      | <b>15.00</b> | -14.00      | <b>15.00</b> | -14.00      | <b>15.00</b> | -0.50 | -6.30 | -17.50      | -0.54      | -        |
| output level [dBmW]                | <b>-5.00</b>   | 17.50        | 3.00        | 17.50        | 3.00        | 17.50        | 3.00        | 17.50        | 17.00 | 10.70 | -6.80       | -7.34      | RF80dBuV |
| output $R/N$ [dB/Hz <sup>1</sup> ] | <b>-159.10</b> | -152.73      |             | -149.46      |             | -147.62      |             | -146.33      | ←     | ←     | ←           | ←          |          |
| channel C/N [dB]                   |                |              |             |              |             |              |             |              | ←     | ←     | ←           | ←          | 48.55    |
| system C/N [dB]                    |                |              |             |              |             |              |             |              | ←     | ←     | ←           | ←          | 46.55    |

NOTE 1 optical channel C/N at the V-ONU RF output, which is calculated from *R/N* of the optical transmitter and the amplifier  
NOTE 2 C/N at V-ONU RF output, which is calculated from the receiving point C/N, transmitter C/N and optical channel C/N  
NOTE 3 connection loss between optical amplifier and connection panel is included in the loss of the connection panel

**Figure A.8 – Model No.4 of a system performance calculation**

**Model No.5 – Multi-channel service system: 4 steps of EDFA(s); OMI: analogue 7 % (11 carriers), digital 2,2 % (80 carriers) with a WDM filter**



| item                            | value   |
|---------------------------------|---------|
| amplifier NF                    | 6,2 dB  |
| OMI/ch                          | 7,0 %   |
| receiving point C/N             | 53,0 dB |
| transmitterB <sub>out</sub> C/N | 55,0 dB |

[illegible]

NOTE 1 optical channel C/N at the V-ONU RF output, which is calculated from  $R/N$  of the optical transmitter and the amplifier

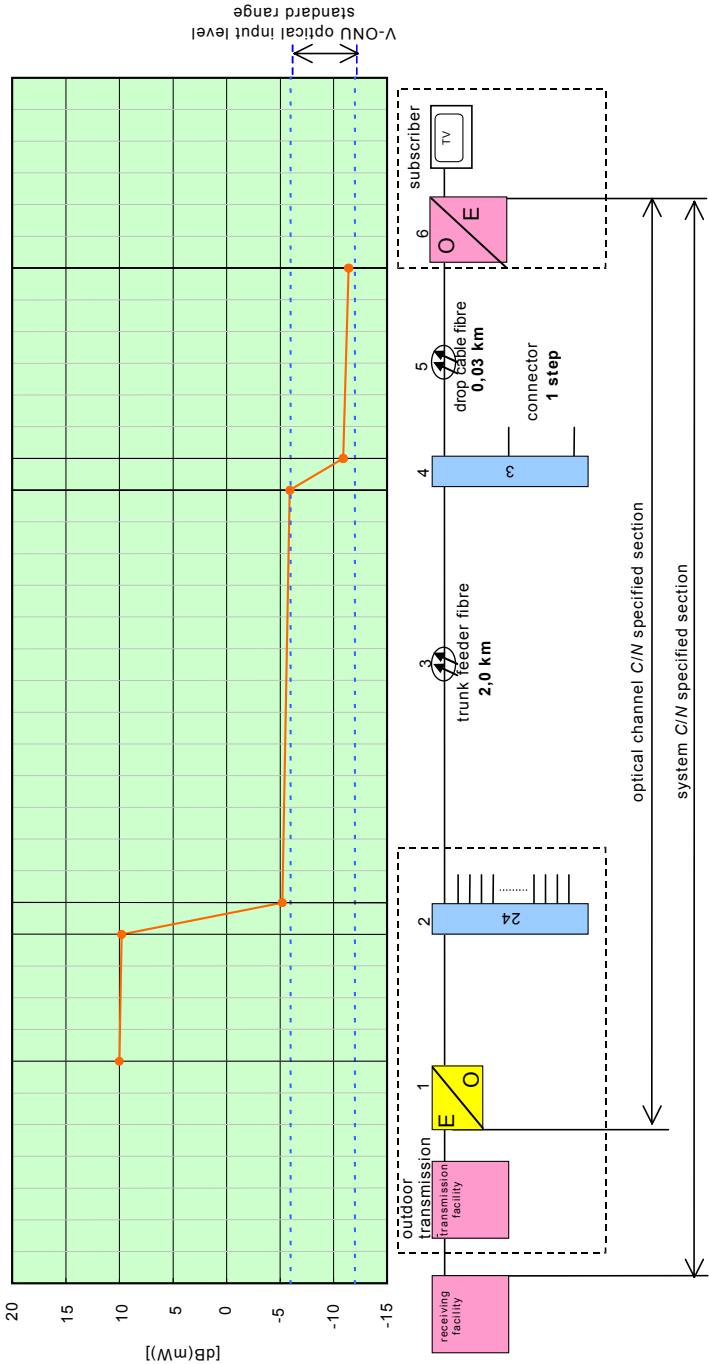
NOTE 1 Optical channel C/N at the V-ONU RF output, which is calculated from the optical transmitter and the amplifier.

NOTE 2 C/N at V-ONU RF output, which is calculated from the receiving point C/N, transmitter C/N and optical channel C/N

NOTE 3 connection loss between optical amplifier and connection panel is included in the loss of the connection panel

**Figure A.9 – Model No.5 of a system performance calculation**

Model No.6 – Re-transmission service system with no EDFA; OMI: analogue 9 % (9 carriers), digital 2,9 % (9 carriers)



| Item                | value   |
|---------------------|---------|
| OMI /ch             | 9,0 %   |
| receiving point C/N | 48,0 dB |
| transmitter RF out  | 55,0 dB |

| equipment                          | 1           | 2      | 3          | 4          | 5      | 6         |
|------------------------------------|-------------|--------|------------|------------|--------|-----------|
| transmitter                        | 24 splitter | trunk  | 3 splitter | drop cable | V-ONU  |           |
| input level [dBmW]                 | 9,80        | -5,20  | -5,90      | -10,90     | -11,41 |           |
| gain or loss [dB]                  | -           | -15,00 | -0,70      | -5,00      | -0,51  |           |
| output level [dBmW]                | 10,00       | -5,20  | -5,90      | -10,90     | -11,41 | RF:80dBuV |
| output R/N [dB(Hz <sup>-1</sup> )] | -150,00     | ←      | ←          | ←          | ←      |           |
| channel C/N [dB]                   | -           | -      | -          | -          | -      | 46,59     |
| system C/N [dB]                    | -           | -      | -          | -          | -      | 43,88     |

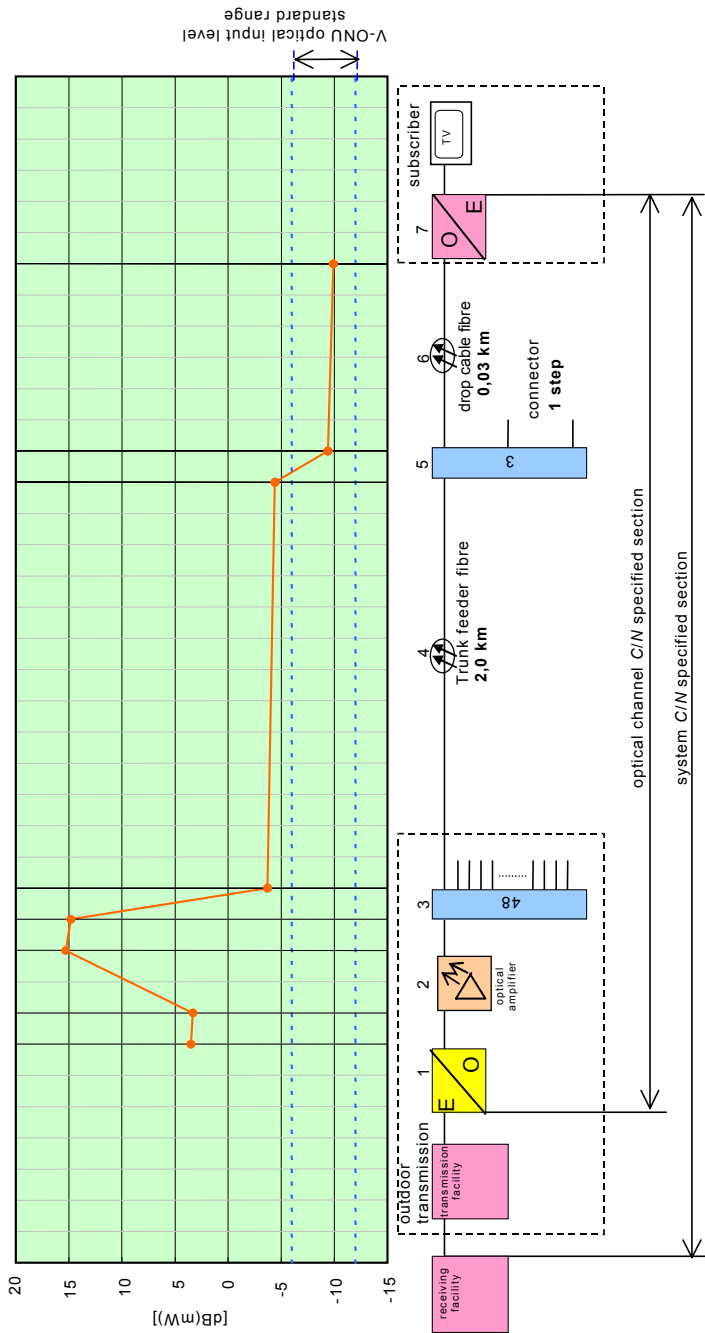
NOTE 1 optical channel C/N at the V-ONU RF output, which is calculated from R/N of the optical transmitter and the amplifier

NOTE 2 C/N at V-ONU RF output, which is calculated from the receiving point C/N, transmitter C/N and optical channel C/N

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Figure A.10 – Model No.6 of a system performance calculation

Model No.7 – Re-transmission service system 1 step of EDFA; OMI: analogue 9 % (9 carriers), digital 2,9 % (9 carriers)



| item                | value   |
|---------------------|---------|
| amplifier NF        | 6,2 dB  |
| OMI/ch              | 9,0 %   |
| receiving point C/N | 48,0 dB |
| transmitter RF out  | 55,0 dB |

| equipment                          | 1         | 2         | 3           | 4     | 5          | 6          | 7         |
|------------------------------------|-----------|-----------|-------------|-------|------------|------------|-----------|
| transmitter                        | RF:80dBuV | amplifier | 48 splitter | trunk | 3 splitter | drop cable | V-ONU     |
| input level [dBmW]                 |           | 3,30      | 14,80       | -3,70 | -4,40      | -9,40      | -9,91     |
| gain or loss [dB]                  |           | 12,00     | -18,50      | -0,70 | -5,00      | -0,51      | -         |
| output level [dBmW]                |           | 15,30     | -3,70       | -4,40 | -9,40      | -9,91      | RF:80dBuV |
| output RIN [dB(Hz <sup>-1</sup> )] |           | -150,00   | -148,24     |       |            |            |           |
| channel C/N [dB]                   |           |           |             |       |            |            | 48,87     |
| system C/N [dB]                    |           |           |             |       |            |            | 44,95     |

NOTE 1 optical channel C/N at the V-ONU RF output, which is calculated from R/N of the optical transmitter and the amplifier  
NOTE 2 C/N at V-ONU RF output, which is calculated from the receiving point C/N, transmitter C/N and optical channel C/N

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Figure A.11 – Model No.7 of system performance calculation

## A.5 Tips for actual operation

### A.5.1 Optimum operation

This clause describes neither a system performance nor device parameters. They may be optimized with the number of transmit channels as long as they satisfy the allowable ranges of *RIN*, *NF*, *C/N* ratio and the interferences described in this standard. Table A.4 shows an example of parameters verified in actual broadcast signal transmission. As long as the system performance is maintained, the operation with other modulation indexes can be allowed. Even if the modulation index is the same in this table and the number of carriers differs, the system is still considered to be satisfactory as long as the *RIN* value, the number of carriers and the *C/N* ratio at V-ONU output are maintained in adequate condition.

**Table A.4 – Verified optimum operation**

| Service type                                                                                                                              | Analogue multi-channel system      |                |                                 |     | Digital multi-channel system       |     |                 |     | Re-transmission system             |     |                 |     |
|-------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------|----------------|---------------------------------|-----|------------------------------------|-----|-----------------|-----|------------------------------------|-----|-----------------|-----|
|                                                                                                                                           | Combination (1)<br>ref. model 1, 2 |                | Combination (2)<br>ref. model 3 |     | Combination (3)<br>ref. model 4, 5 |     | Combination (4) |     | Combination (5)<br>ref. model 6, 7 |     | Combination (6) |     |
| Number of carriers                                                                                                                        | A <sup>a</sup>                     | D <sup>a</sup> | A                               | D   | A                                  | D   | A               | D   | A                                  | D   | A               | D   |
|                                                                                                                                           | 57                                 | 40             | 40                              | 30  | 11                                 | 80  | -               | 110 | 9                                  | 9   | 10              | 10  |
| Modulation index<br>% $m_k$                                                                                                               | 3,5                                | 1,1            | 4,3                             | 1,4 | 7                                  | 2,2 | -               | 2,7 | 9                                  | 2,9 | 8,6             | 2,7 |
| <sup>a</sup> A = analogue; D = digital.                                                                                                   |                                    |                |                                 |     |                                    |     |                 |     |                                    |     |                 |     |
| NOTE Combinations (1), (2), (3) and (5) correspond to the reference model of Clause A.4. Combinations (4) and (6) are for reference only. |                                    |                |                                 |     |                                    |     |                 |     |                                    |     |                 |     |

### A.5.2 Key issues to be specified

The key issues in order to design an optical system and operating conditions are shown below.

#### a) Optical wavelength

Although the wavelength of 1 530 nm to 1 625 nm is generally used, many commercially available systems use wavelengths of 1 540 nm to 1 560 nm. This band (i.e. 1 540 nm – 1 560 nm) is therefore chosen to limit the operating wavelength. Nevertheless, with the progress of WDM technology usage of wavelengths may vary.

#### b) Optical modulation system

As the optical modulation system in CATV network, only the intensity modulation system is specified in this standard. The re-transmission service system selects a direct intensity modulation system due to its short trunk line length, and a multi-channel service system



introduces an external intensity modulation system considering its long span ( approximately 40 km at maximum). In case of short span in a multi-channel service system, direct intensity modulation can also be utilized.

c) Optical modulation index

Optical modulation index in some system models are described in Table A.4. Many optical transmitters in the market have the function to adjust the modulation index automatically according to the number of carriers by its AGC function. As mentioned in A.5.1, the operation with other adequate modulation indexes can be allowed as long as system performance is satisfied.

d) Distortion performance

Since it is difficult to describe system performance separately between optical transmitter and V-ONU, it is recommended to specify the performance with a pair of optical transmitter and V-ONU. The distortion performance of optical system may change its optical characteristics in actual operation. Especially, this factor is significant when direct intensity modulation is used in re-transmission system. Careful selection of each distortion parameter is required for actual optical system design.

e) V-ONU optical input level

This specification does not limit the operation beyond parameters described in this document. It specifies the minimum range of specification. Hence, this specification can be used as a guideline of system design. The specified optical input level is assumed without WDM filters, the loss shall be compensated if it has WDM filters.

f) AGC function of V-ONU

There is a V-ONU having AGC function that detects optical input level and controls electric power sum of the output signal. This standard does not specify this type of control, but requires to keep the output level within the fixed range even if the optical input level of V-ONU varies.

g) *C/N* ratio (*CNR*: *C/N*)

The *C/N* ratio in CATV network generally shows lowest value at optical input of V-ONU. It is possible to increase input optical level to some extent to meet higher *C/N* ratio requirement.

## Annex B (informative)

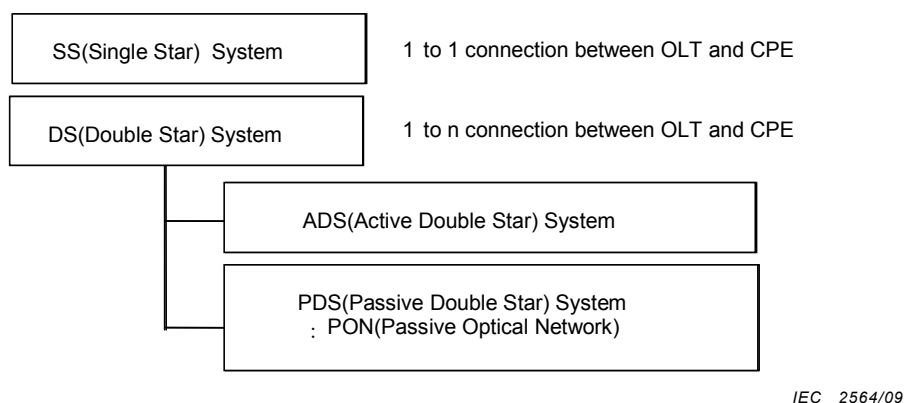
### Optical system overview

#### B.1 Optical system topology

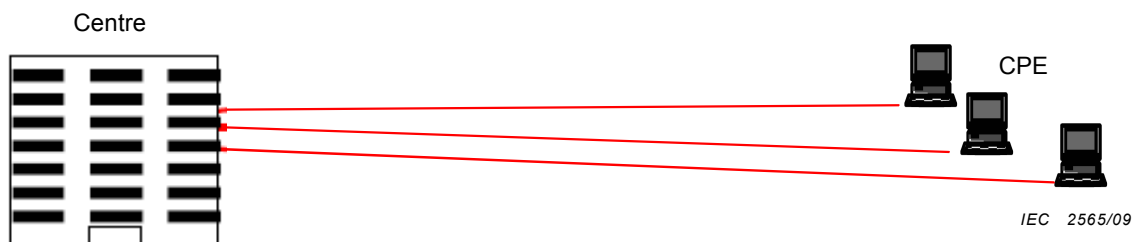
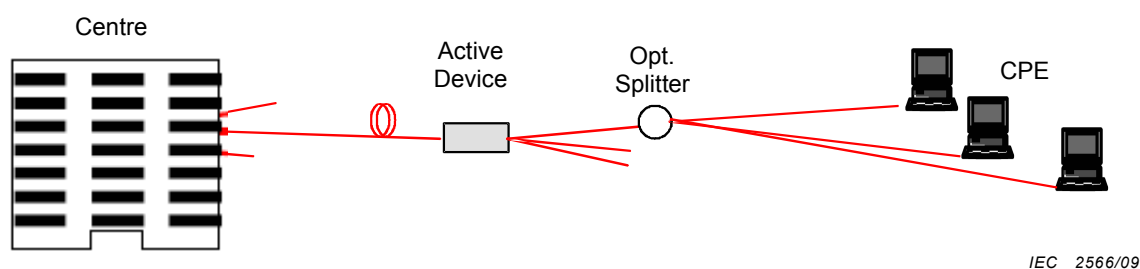
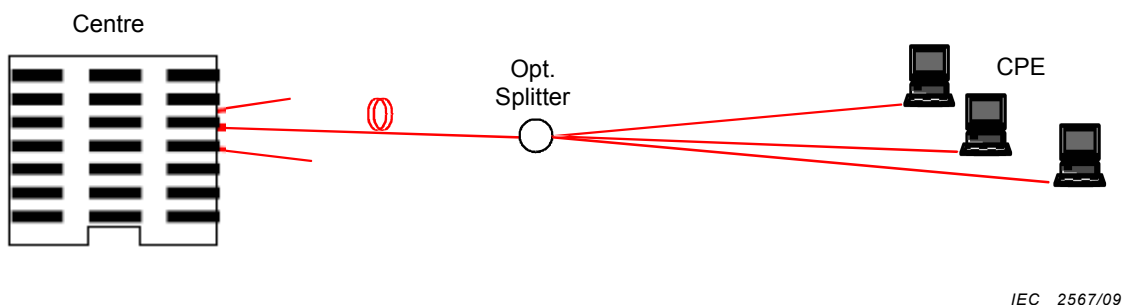
##### B.1.1 Single star and double star system

This annex describes the optical system overview that is applicable to broadcast signal transmission.

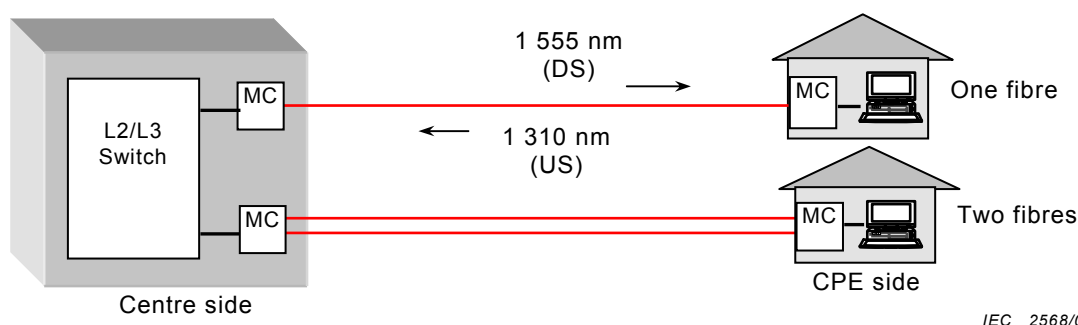
The optical network is classified into Single Star (SS) and Double Star (DS) systems. An active Double Star (ADS) or Passive Double Star (PDS) system is a kind of DS system. PDS is also called Passive Optical Network (PON). The progress of wavelength division multiplex (WDM) technology allows its application to video signal transmission. Figure B.1 shows the topology of the optical system and Figure B.2 the network composition.



**Figure B.1 – Topology of optical system**

**Figure B.2a – SS system****Figure B.2b – ADS system****Figure B.2c – PDS system****Figure B.2 – Network composition****B.1.2 SS system**

Single Star is a system containing one central terminal and one customer premises equipment (CPE) in a pair. Media Converter (MC) is installed in each centre office and CPE, and it converts an electrical signal to an optical signal and vice versa. There is a component type for individual CPE, and a consolidated type which has slots for media converter modules. It has also maintenance, failure detection and provisioning functions. High-speed services such as tenth of Mbps to Gbps are available. One or two optical fibres are used for this system. An example of an SS system is shown in Figure B.3.

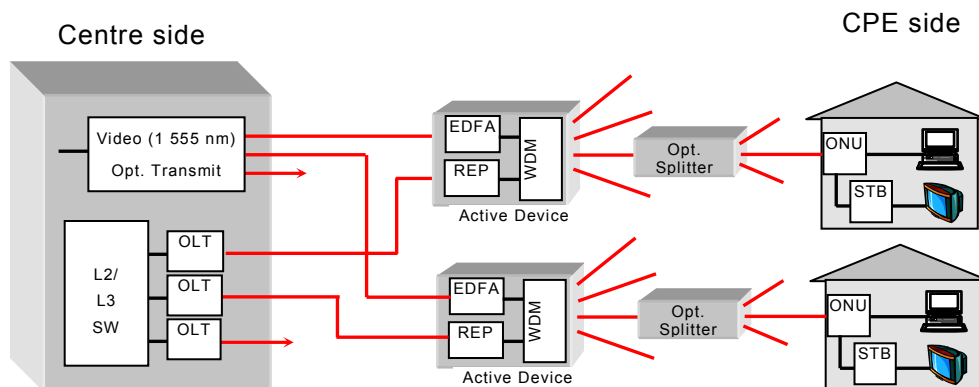


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**Figure B.3 – Example of SS system**

### B.1.3 ADS system

The ADS system is used when the broadband distribution network is required in a wide area. It consists of optical fibre, splitter, optical amplifier, WDM and LAN switches. An optical fibre is distributed through the optical splitter installed near the customer premises. Video and other data occupies fibre individually between the centre and an active device. In the active device wavelengths for video and other data are multiplexed onto one fibre, then transmitted to splitters. Other splitters are provided to distribute fibres to subscribers. The end section is a dropping fibre to CPE. 1 555 nm wavelength is used for video, 1 490 nm (DS) and 1 310 nm (US) wavelengths are applied for data transmission. A 1 310 nm wavelength system for both DS and US is also commercialized. ADS has the active devices in the transmission line, which requires electric power, and this may impair the system reliability. The system reliability should be improved by inserting a redundancy network between the centre and the CPE, and by providing a status monitoring and control system. An example of an ADS system is shown in Figure B.4.

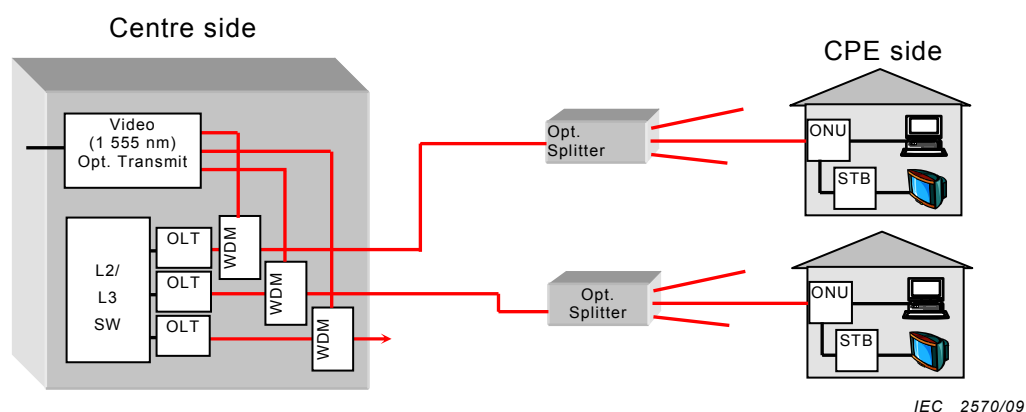


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**Figure B.4 – Example of ADS system**

### B.1.4 PON system

A PON system uses only passive elements (optical splitters) for the branch section, and constitutes an access network. Since the optical line from the centre to the optical splitter is shared by multiple subscribers, the cost of the optical-fibre construction can be reduced. Due to using a non-active device the PON system can provide a high-reliable network. PON can also offer simple monitoring and facilitate maintenance for cable operators. There are G-PON of gigabit class, GE-PON by the wavelength multiplex, and B-PON of 622/156 Mbps classes. A three-wavelength multiplex system is commercialized. It uses 1 555 nm for video and 1 310 nm/1 490 nm for US/ DS of data. An example of a PON system is shown in Figure B.5, and G-PON, GE-PON and B-PON systems are given in Table B.1.

**Figure B.5 – Example of PON system****Table B.1 – PON systems and main parameters**

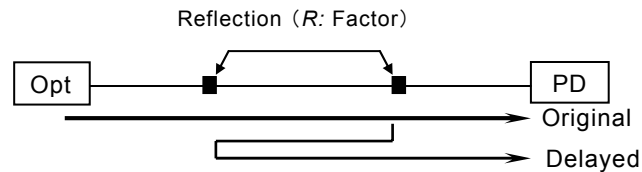
| Standard common-name       |       | G-PON                                               | GE-PON               | B-PON                |
|----------------------------|-------|-----------------------------------------------------|----------------------|----------------------|
| Frame                      |       | Generic frame (ATM/STM/Ethernet)                    | Ethernet frame       | ATM frame            |
| Wave-length                | US    | 1260-1360 nm                                        | 1 260 nm to 1 360 nm | 1 260 nm to 1 360 nm |
|                            | DS    | 1480-1500 nm                                        | 1 480 nm to 1500 nm  | 1 480 nm to 1 500 nm |
|                            | Video | 1550-1560 nm                                        | 1 550 nm to 1560 nm  | 1 550 nm to 1 560 nm |
| Bit Rate                   | DS    | 1,25G-2,5 Gbit/s                                    | 1,25 Gbit/s          | 622 Mbit/s           |
|                            | US    | 1,25 Gbit/s                                         | 1,25 Gbit/s          | 156 Mbit/s           |
| The number of PON branches |       | 64 (Logical: 128)                                   | 16 or over           | 32                   |
| Transmission distance      |       | 10/20 km<br>Theoretical: 1 550 nm to 1 560 nm 60 km | 10/20 km             | 20 km                |

## Annex C (informative)

### Optical system degradations

#### C.1 System degradation factors

The transmission quality in telecommunication can be evaluated by *BER* for digital signal, however, in case of broadcast signal that contains a mixture of analogue and digital signals, the performance evaluation may include *C/N* ratio, *BER*, *MER*, signal distortion and subjective picture quality. Subjective picture quality evaluation is outside of scope of this standard. Generally, a broadcast system must also consider various non-linear degradation factors due to higher optical level usage than telecommunication system. Furthermore, in case of the three-wavelength multiplex transmission system, no interference among services (video and data) is strictly required. In video transmission, the optical reflection over transmission line should be specifically considered as a keen degradation factor. Relative Intensity Noise (*RIN*) is used as an index for evaluation of noise property of optical signal. Although *RIN* is the parameter showing time fluctuation of laser power, it can evaluate noise characteristics caused by optical reflection in the system. The model that has two reflection points in a transmission line is shown in Figure C.1. A beat arises between the twice-reflected light (delayed signal) and direct signal (original signal), and this serves as noise.



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**Figure C.1 – Reflection model**

When LD is directly modulated by the multi-channel signal, *RIN* is expressed as follows.

$$RIN_{\text{ref}(f)} = \frac{2\alpha R_1 R_2}{\sqrt{\pi \sigma_f}} \cdot \exp \left[ -f^2 / 4\sigma_f^2 \right] \quad (\text{C.1})$$

where

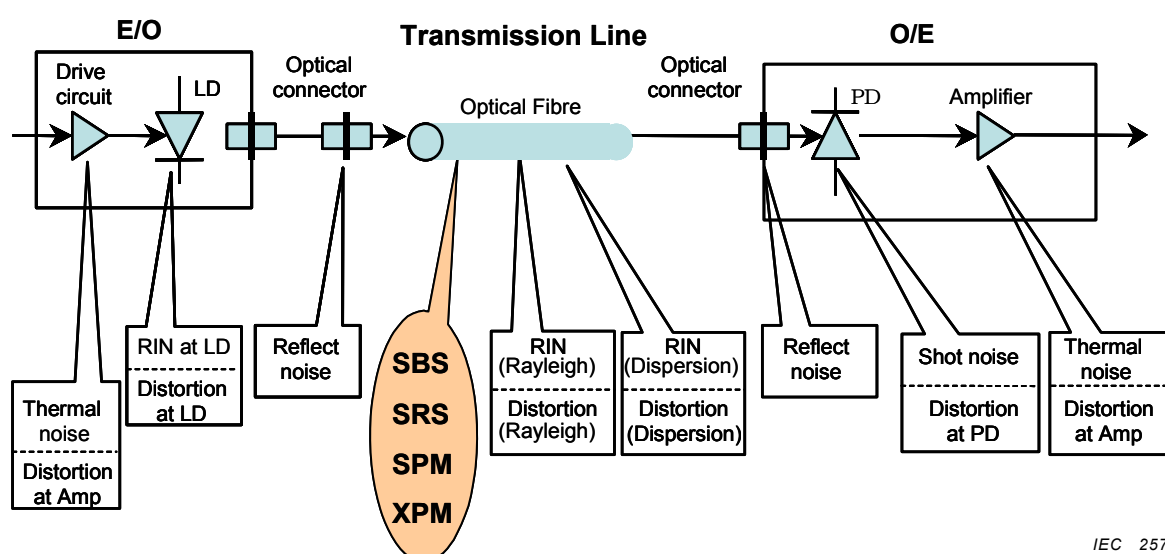
- $R_1, R_2$  is the reflective index for first and second time,
- $\alpha$  is the optical loss between two reflection points,
- $\sigma_f$  is the optical frequency shift of LD.

On a transmission line, reflection may occur at the point of the fibre connection or the connector. For example, if in the observing point  $\sigma_f$  is set to 1,3 GHz,  $\alpha$  to 5 dB,  $R_1$  and  $R_2$  to 40 dB each, in case of normal Grade 2 connectors of IEC 61755-1 polish, the *RIN* obtained at  $f = 100$  MHz will be  $-175$  dB(Hz<sup>-1</sup>). If 20 dB is similarly substituted for  $R_1$  and  $R_2$ , in case of PC polish, *RIN* will be  $-135$  dB(Hz<sup>-1</sup>). Therefore, in this case the system will be deteriorated significantly. Generally, since *RIN* of an optical laser is about  $-160$  dB(Hz<sup>-1</sup>), the former Grade 2 connectors of IEC 61755-1 polish case may not cause interference, while the latter PC polish case will affect the system performance. Therefore, in the case of CATV, the Grade 2 connectors of IEC 61755-1 is recommended, and it is better to use an angled-PC polish type having reflection loss of 60 dB or more.

## C.2 Non-linear degradation

### C.2.1 Degradation factors

Various degradation factors over an optical transmission system are described in Figure C.2. The nonlinear degradation factor produced in the optical fibre itself has been causing attention in recent years. Important parameters are stimulated Brillouin scattering (SBS), stimulated Raman scattering (SRS), self-phase modulation (SPM), cross phase modulation (XPM). These types of degradation are caused by a high-level optical power input into optical fibres. It is not enough to specify these parameters only for the optical transmitter or the optical receiver. As a total cable system, it is necessary to design a complete system with the appropriate optical level, as described in Annex A.

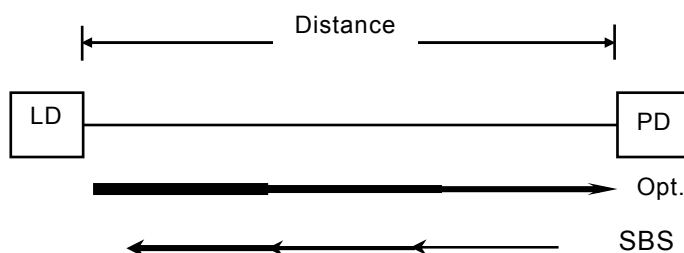


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Figure C.2 – Degradation factors of optical transmission system

### C.2.2 Stimulated Brillouin Scattering (SBS)

SBS is a scattering effect of silica fibre due to its non-linear characteristics and caused by an excessive optical input power over some threshold. In this case dispersion light cannot reach the receiving side but return to the sending side. Figure C.3 shows an SBS generation image. In order to expand the threshold level range, a relaxing method of energy density is used by applying frequency modulation or phase modulation to the optical source of the transmitter. Currently a +20 dB(mW) power class is available.



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Figure C.3 – SBS generation image

### C.2.3 Stimulated Raman scattering (SRS)

This is an interference phenomenon from telecommunication signals to video signals (from the short wavelength to the long wavelength side) due to stimulated Raman scattering, a non-linear optical effect of optical fibre. The level of interference depends on the optical input level (launch power), transmission distance, wavelength distance and the spectrum of telecommunication signals. As the Raman scattering has its peak gain at 100 nm apart from the original wavelength on the long wavelength side in case of silica fibre, the interference occurs from the short wavelength side (telecommunication signals) to the long wavelength side (video signals). Table C.1 shows the tendency of SRS disturbance with its main cause.

**Table C.1 – Disturbance parameter of Raman crosstalk**

| Main parameters                             | Tendency of SRS disturbance                                                                                                |
|---------------------------------------------|----------------------------------------------------------------------------------------------------------------------------|
| Optical-fibre input power (launch power)    | SRS increases in accordance with launch power.                                                                             |
| Fibre length                                | SRS increases in accordance with fibre length.                                                                             |
| Wavelength interval                         | Raman gain peak is about 100 nm on the long wavelength side.                                                               |
| Frequency of CATV                           | Interference is likely to occur on the low frequency side.                                                                 |
| Optical modulation index<br>In video signal | Interference is likely to occur in the low optical modulation condition and improves by increment of the modulation index. |
| Kind of optical fibre                       | DSF is more likely to suffer from interference than SMF.                                                                   |
| Spectrum of telecommunication signal        | The dispersed spectrum component (in idling state) may suffer from interference.                                           |

The CCR (carrier-to-crosstalk ratio) effect on the CATV signals is due to the Raman scattering effect. The CCR effect can be evaluated by the formula shown below:

$$CCR_{SRS} = 10 \lg \left\{ \left[ \frac{A_{\text{eff}}}{\rho_{SRS} g_{12} P_{\text{int}}} \cdot \frac{m_{\text{CATV}}}{m_{\text{int}}} \right]^2 \cdot \frac{\alpha^2 + (\Omega d_{12})^2}{1 + e^{-2\alpha L} - 2e^{-\alpha L} \cos(\Omega d_{12} L)} \right\} \quad (\text{C.2})$$

where

$A_{\text{eff}}$  is the effective area,

$\rho_{SRS}$  is the effective polarization overlap factor,

$g_{12}$  is the Raman gain coefficient,

$P_{\text{int}}$  is the optical power of the interfering signal,

$m_{\text{CATV}}$  is the optical modulation index (OMI) of CATV signal,

$m_{\text{int}}$  is the OMI of the interfering signal,

$\alpha$  is the attenuation coefficient of the fibre,

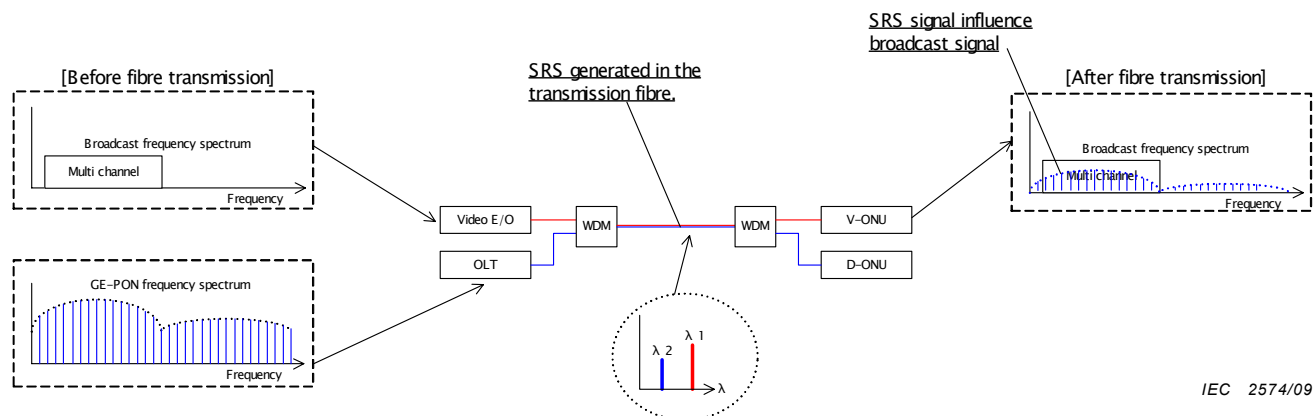
$d_{12}$  is the group velocity mismatch between two WDM signals,

$L$  is the fibre length.

As shown in this formula, CCR can be improved by increasing modulation index of CATV signal or decreasing signal level and modulation index of telecommunication signals. The  $D/U$  (desired to undesired signal level) ratio degrades as a cosine curve in accordance with the fibre length.

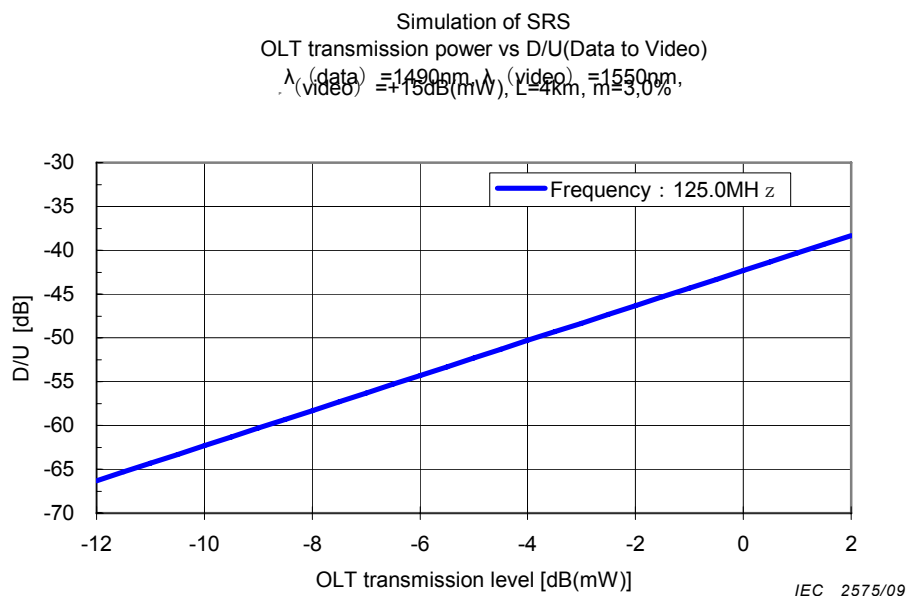


In a hybrid WDM transmission system in which both the broadcast signal and the telecommunication signal are incorporated, interference between the two signals due to optical fibre non-linearity shall be taken into consideration. Figure C.4 shows the interference between two wavelengths;  $\lambda_1$  for optical broadcast signal and  $\lambda_2$  for optical telecommunication signal. Generally, Raman gain peak arises at about 100 nm on the long wavelength side in a silica fibre. The interference from the telecommunication signal (digital) to the broadcast signal (analogue) must be considered. As, normally,  $\lambda_1$  is set to 1 490 nm and  $\lambda_2$  to 1 550 nm in a hybrid WDM system, SRS becomes a dominant interference between the two signals.



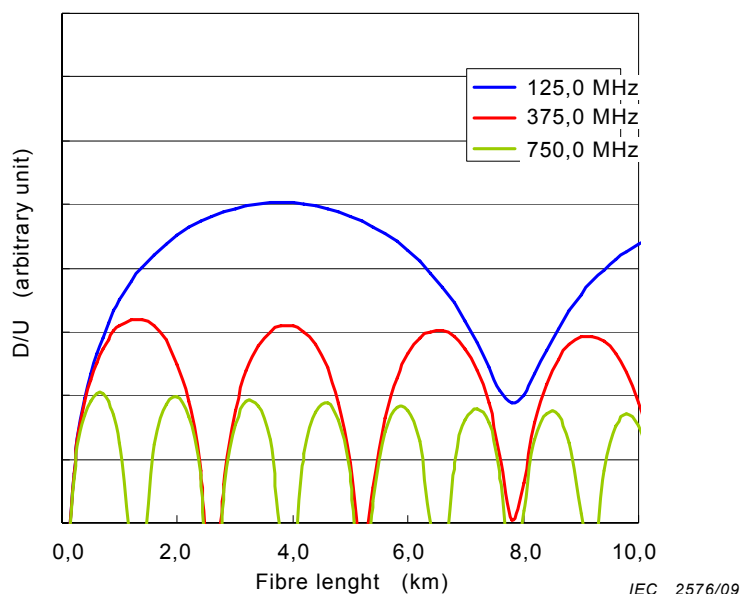
**Figure C.4 – Interference between two wavelengths**

SRS occurs in a fibre where two wavelengths are incorporated, and depends on the parameters such as the length of fibre, optical input levels, polarization of optical signals and transmission frequencies.



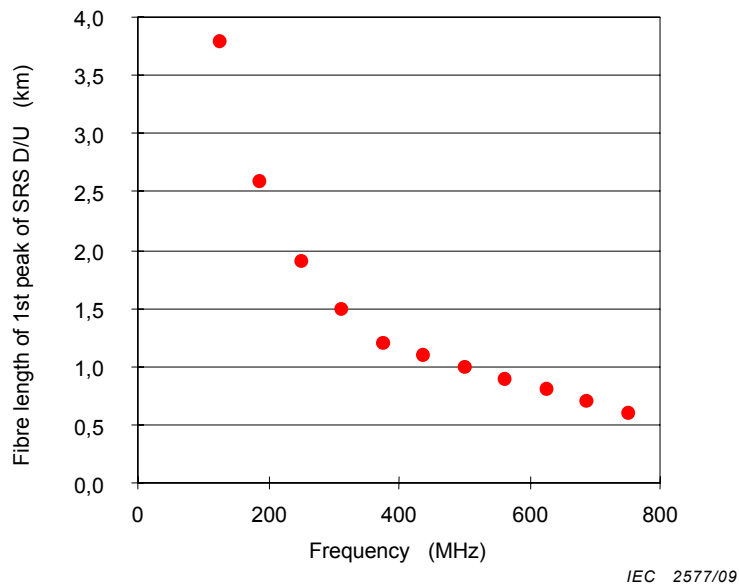
**Figure C.5 – Simulation of SRS(OLT transmission power versus D/U)**

As shown in Figure C.5, SRS depends on the optical input level of OLT (1 490 nm). It is noted that the optical input level of 1 490 nm wavelength shall be kept between -7 dB(mW) to 0 dB(mW) if the level difference between the two signals is 18 dB.



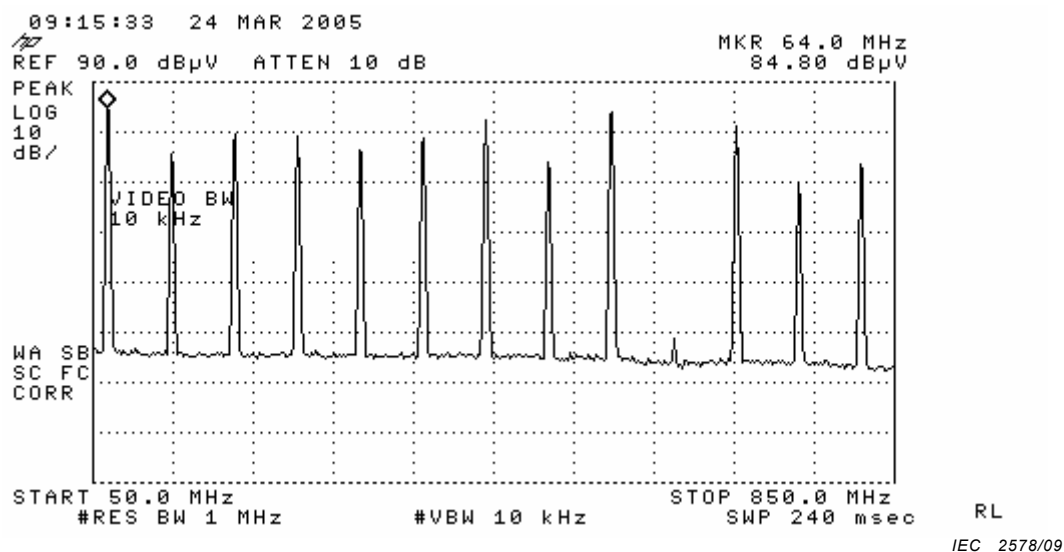
**Figure C.6 – Simulation of SRS (D/U in arbitrary unit versus fibre length)**

Figure C.6 shows a relationship between SRS and the fibre length. It is noted that each peak of SRS has a cosine curve and it has strong dependency with signal frequencies, not fibre length.



**Figure C.7 – Fibre length of the first peak of SRS D/U versus frequency**

Figure C.7 shows a relationship between the highest SRS D/U and signal frequency. For example, the 3.9 km fibre length point indicates the worst SRS D/U at a signal frequency of 125 MHz.



**Figure C.8 – GE-PON idle pattern spectrum (IEEE 802.3ah 1000Base-PX)  
 (62,5 MHz = 1 250 Mbps/20 bit)**

Figure C.8 shows the spectrum of GE-PON in idling condition, where periodic high energy spectrums arise. In actual transmission this spectrum is randomized by data signal and hence the peak power is lowered. It is noted that some data randomization is useful for dispersing the spectrum.

#### **C.2.4 Self-phase modulation (SPM)**

*SPM* is a kind of phase modulation due to the refractive index change caused by optical input power.

#### **C.2.5 Cross-phase modulation (XPM)**

As *SRS*, *XPM* occurs in the wavelength division multiplex system. It is caused by excessive optical power into one optical fibre that draws a refractive index change and this may cause a phase modulation in another optical signal in the same optical fibre. *XPM* occurs rather on the high frequency side of the CATV frequency band that occupies the digital broadcast band. However, the effect is negligible if the wavelength distance is set to be apart.

Although *SBS*, *SRS*, *SPM* and *XPM* are degradations due to the nonlinear effect in the fibre caused by the optical input power, the influence on the whole system performance can be minimized if the optical signal level and wavelength interval are appropriately selected.

## Annex D (normative)

### Measurement of parameters ( $R$ , $I_{d0}$ , $I_{eq}$ and $G$ ) required for $R/N$ calculation

#### D.1 Measurement of responsivity ( $R$ )

The responsivity ( $R$ ) can be measured using the following procedure.

- A CW optical signal (un-modulated) shall be applied to the optical receiver setup.
- Measure the response of the photodiode using the following procedure:
  - adjust the optical attenuator to an optical level of +1 dB(mW) ( $P_{+1}$ ), measured with the optical power meter;
  - after replacing the optical power meter by the optical receiver, measure the current ( $I_{+1}$ ) of the photodiode using the current meter;
  - adjust the optical attenuator to obtain an optical input power of –1 dB(mW) ( $P_{-1}$ ) and measure the corresponding current ( $I_{-1}$ ).
- The responsivity ( $R$ ) of the photodiode at 0 dB(mW) optical input can be calculated using the following equation:

$$R = \frac{I_{+1} - I_{-1}}{\frac{P_{+1}}{10} - \frac{P_{-1}}{10}} = \frac{I_{+1} - I_{-1}}{1,259 - 0,794} = \frac{I_{+1} - I_{-1}}{0,465} \quad [\text{A/W}] \quad (\text{D.1})$$

#### D.2 Measurement of dark current ( $I_{d0}$ )

The dark current of the photodiode can be measured using the following procedure:

- turn off the optical input to the photodiode;
- under the normal biasing condition of the photodiode, measure the current ( $I_{d0}$ ) through the photodiode.

NOTE The dark current is usually negligibly small and can be ignored in the calculation of  $R/N$ .

#### D.3 Measurement of equivalent noise current density ( $I_{eq}$ )

The equivalent optical receiver input noise shall be measured using the following procedure.

- Step a: While turning off the optical input under normal biasing condition of the photodiode, measure the output noise power of the receiver using the  $NF$  meter. In general, the equivalent optical input noise is frequency dependent. Therefore, it is recommended to repeat this measurement for at least 3 different frequencies,  $f_{\min}$  at the lowest transmission frequency,  $f_{\max}$  at the highest transmission frequency and  $f_{\text{mid}}$  in the middle of the transmission frequency range. The noise figure meter will display 3 noise figures for these 3 frequencies,  $NF_{\min}$ ,  $NF_{\text{mid}}$  and  $NF_{\max}$ .
- Step b: A CW optical signal shall be applied to the optical receiver setup via the optical attenuator. It is important to use a CW source with a very low value of  $R/N$  which is lower than  $-155 \text{ dB}(\text{Hz}^{-1})$  in order to eliminate an impact of the  $R/N$  of the CW source on the next measurement. Otherwise the  $R/N$  of the CW source has to be deducted from the noise measurement.

- Step c: The optical attenuator is adjusted to obtain a reading of the  $NF$  meter which is the reading of the measurement from step a + 3 dB, for example  $NF = NF_{\min} + 3 \text{ dB}$  for measuring the output noise at the lowest frequency.
- The 3 dB increase of the  $NF$  meter reading means that the output noise of the receiver comprises the shot noise of the photodiode and exactly the same amount of noise originating from the thermal noise of the receiver,  $N_{\text{shot}} = N_{\text{thermal}}$ , and corresponds to  $I_{\text{sh}} = I_{\text{eq}}$ .

NOTE This relationship is only true as long as the  $RIN$  of the source can be neglected.

- Step d: The shot noise current  $I_{\text{sh}}$  of the photodiode can be easily calculated after measuring the photodiode current:  $I_{\text{dc}}$

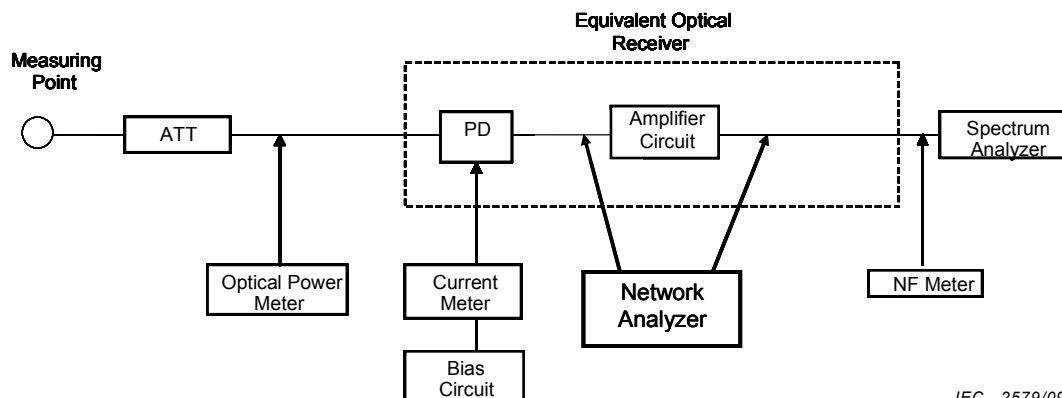
$$I_{\text{eq}} = I_{\text{sh}} = \sqrt{2 \cdot I_{\text{dc}} \cdot e} \quad [A/\sqrt{\text{Hz}}] \quad (\text{D.2})$$

where

$I_{\text{eq}}$  is the amplifier equivalent input noise current density ( $A/\sqrt{\text{Hz}}$ ),  
 $I_{\text{sh}}$  is the photodiode shot noise current density ( $A/\sqrt{\text{Hz}}$ ),  
 $I_{\text{dc}}$  is the photodiode bias current (A),  
 $e$  is the electron charge =  $1,602 \times 10^{-19}$  (C).

#### D.4 Measurement of gain (G)

Use a network analyzer for the measurement of the amplifier circuit gain (G) as shown in Figure D.1.



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Figure D.1 – Measurement of gain (G)

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